

Pre-recorded Tasks Visual Inspection Report

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Contents

1	Overview	2
2	Catalogue Tasks	3
2.1	Summary	3
2.2	Detailed Task Examples	3
2.2.1	Sorted_by_rank	3
2.2.2	S_before_H	4
2.2.3	Ends_same_suit	4
2.2.4	Ends_same_color	4
2.2.5	Has_pair_ranks	5
2.2.6	Uniform_color	5
2.2.7	Exactly_two_suits	6
2.2.8	Half_or_more_same_suit	6
2.2.9	At_most_three_suits	6
2.2.10	Exactly_one_club	7
2.2.11	AP_len3_anywhere_anyk	7
2.2.12	AP_len3_step2_anywhere	7
2.2.13	AP_len4_step2_anywhere	8
2.2.14	Half_sum_diff_geN	8
2.2.15	Half_sum_one_side_ge_2x_other	8
2.2.16	Halves_uniform_color_equal	9
2.2.17	Halves_AP_step1_equal	9
2.2.18	Halves_hearts_presence_equal	9
2.2.19	Halves_AP_len3_any_equal	10
2.2.20	Halves_AP_len2_step1_equal	10
2.2.21	Well_formed_brackets_by_suit	10
2.2.22	Even_opens_next_closes	11
2.2.23	Odd_opens_next_closes	11
2.2.24	Suits_palindrome	11
2.2.25	Colors_palindrome	12
2.2.26	Ranks_palindrome	12
2.2.27	AltColor1_palindrome	12
2.2.28	AltColor2_palindrome	13
2.2.29	Ends_same_altcolor1	13
2.2.30	Halves_copy_suits	13
2.2.31	Halves_copy_colors	14
2.2.32	Halves_copy_ranks	14
2.2.33	Halves_copy_altcolor1	14
2.2.34	Halves_copy_altcolor2	15
2.2.35	Halves_same_suit_set	15
2.2.36	Shift_half_ge	15
2.2.37	Adj_same_rank_or_suit	16
2.2.38	Skip2_same_rank_or_suit	16
2.2.39	Adj_rank_gap_le3	16
2.2.40	Adj_same_rank_or_color	17

2.2.41	Only_one_odd_rank	17
2.2.42	Uniform_rank_parity	17
2.2.43	Halves_radial_nonincreasing	18
2.2.44	Global_radial_no_dominance	18
3	Pretraining Tasks	19
3.1	Summary	19
3.2	Detailed Task Examples	19
3.2.1	poker_has_pair	19
3.2.2	poker_three_of_kind	20
3.2.3	poker_flush	20
3.2.4	poker_same_color	20
3.2.5	poker_two_suits	21
3.2.6	poker_straight	21
3.2.7	bj_sum_even	22
3.2.8	bj_sum_odd	22
3.2.9	rummy_run_3	22
3.2.10	rummy_set_3	23
3.2.11	rummy_all_different	23
3.2.12	rummy_three_ranks	23
3.2.13	sol_alternating	24
3.2.14	sol_v_shape	24
3.2.15	sol_ascending	24
3.2.16	sol_sorted	25
3.2.17	sol_same_suit_seq	25
3.2.18	simple_first_red	25
3.2.19	simple_last_black	26
3.2.20	simple_ends_suit	26
3.2.21	simple_ends_color	26
3.2.22	simple_ends_rank	27
3.2.23	simple_has_spade	27
3.2.24	simple_has_heart	27
3.2.25	simple_all_even	28
3.2.26	simple_all_odd	28
3.2.27	simple_diverse_ranks	28
3.2.28	simple_two_suits	29
3.2.29	sym_suits_palindrome	29
3.2.30	sym_periodic_colors	29
3.2.31	sym_ranks_palindrome	30
3.2.32	sym_halves_suits	30
3.2.33	count_two_red	30
3.2.34	count_more_red	31
3.2.35	count_three_suit	31
3.2.36	count_majority_color	31
3.2.37	comp_left_half_uniform_color	32
3.2.38	comp_right_half_has_pair	32
3.2.39	comp_skip1_same_color	32
3.2.40	comp_shift2_color	33
3.2.41	comp_first_two_same_suit	33
3.2.42	comp_last_two_same_color	33
3.2.43	comp_binary_suit_alt	34
3.2.44	comp_color_pairs	34
4	Generation Metadata	34
4.1	Catalogue	34
4.2	Pretraining	35

1 Overview

This report shows all pre-recorded tasks generated for reproducible experiments. Each task has:

- **Training examples:** What the model sees during training (balanced 50% positive, 50% negative)
- **Holdout examples:** Used for verification (disjoint from training)
- **Near-miss negatives:** Negatives differ from positives by only ONE card

Key guarantees:

1. All tasks have equal positives and negatives
2. Seed positives (used to generate near-miss negatives) are SEPARATE from training positives
3. Holdout is completely disjoint from training

2 Catalogue Tasks

2.1 Summary

Task	Train+	Train-	Hold+	Hold-	Total
Sorted_by_rank	20	20	20	20	80
S_before_H	20	20	20	20	80
Ends_same_suit	20	20	20	20	80
Ends_same_color	20	20	20	20	80
Has_pair_ranks	20	20	20	20	80
Uniform_color	20	20	20	20	80
Exactly_two_suits	20	20	20	20	80
Half_or_more_same_suit	20	20	20	20	80
At_most_three_suits	20	20	20	20	80
Exactly_one_club	20	20	20	20	80
AP_len3_anywhere_anyk	20	20	20	20	80
AP_len3_step2_anywhere	20	20	20	20	80
AP_len4_step2_anywhere	20	20	20	20	80
Half_sum_diff_geN	20	20	20	20	80
Half_sum_one_side_ge_2x_other	20	20	20	20	80
Halves_uniform_color_equal	20	20	20	20	80
Halves_AP_step1_equal	20	20	20	20	80
Halves_hearts_presence_equal	20	20	20	20	80
Halves_AP_len3_any_equal	20	20	20	20	80
Halves_AP_len2_step1_equal	20	20	20	20	80
Well_formed_brackets_by_suit	20	20	20	20	80
Even_opens_next_closes	20	20	20	20	80
Odd_opens_next_closes	20	20	20	20	80
Suits_palindrome	20	20	20	20	80
Colors_palindrome	20	20	20	20	80
Ranks_palindrome	20	20	20	20	80
AltColor1_palindrome	20	20	20	20	80
AltColor2_palindrome	20	20	20	20	80
Ends_same_altcolor1	20	20	20	20	80
Halves_copy_suits	20	20	20	20	80
Halves_copy_colors	20	20	20	20	80
Halves_copy_ranks	20	20	20	20	80
Halves_copy_altcolor1	20	20	20	20	80
Halves_copy_altcolor2	20	20	20	20	80
Halves_same_suit_set	20	20	20	20	80
Shift_half_ge	20	20	20	20	80
Adj_same_rank_or_suit	20	20	20	20	80
Skip2_same_rank_or_suit	20	20	20	20	80
Adj_rank_gap_le3	20	20	20	20	80
Adj_same_rank_or_color	20	20	20	20	80
Only_one_odd_rank	20	20	20	20	80
Uniform_rank_parity	20	20	20	20	80
Halves_radial_nonincreasing	20	20	20	20	80
Global_radial_no_dominance	20	20	20	20	80

2.2 Detailed Task Examples

2.2.1 Sorted_by_rank

Family: LOCAL, Difficulty: 1

Training Examples (first 6):

Hand	Label
TWO-C, TWO-D, FIV-C, FIV-S, NIN-H, FIV-S	FALSE
TWO-S, TWO-S, THR-S, SIX-C, SEV-C, SEV-S	TRUE
KIN-C, TWO-H, TWO-H, THR-H, FIV-C, TEN-C	FALSE
THR-D, THR-C, SEV-S, JAC-S, JAC-H, JAC-S	TRUE
FIV-C, SIX-C, NIN-H, NIN-S, TWO-D, ACE-C	FALSE
FIV-S, FIV-S, FIV-H, TEN-S, ACE-S, KIN-S	FALSE
Holdout Examples (first 4):	
Hand	Label
EIG-S, FIV-D, TEN-H, NIN-D, TEN-S, QUE-D	FALSE
THR-H, THR-S, THR-H, SEV-S, TEN-H, TEN-S	TRUE
TWO-C, TWO-D, SEV-H, TEN-C, ACE-S, ACE-C	TRUE
NIN-H, THR-C, THR-D, KIN-C, ACE-H, JAC-H	FALSE

2.2.2 S_before_H

Family: LOCAL, Difficulty: 1

Training Examples (first 6):

Hand	Label
TWO-S, EIG-H, TWO-C, ACE-H, TEN-H, KIN-S	TRUE
SIX-H, SIX-S, SEV-S, NIN-S, ACE-D, JAC-H	TRUE
EIG-H, KIN-H, SEV-D, TEN-S, THR-D, JAC-S	FALSE
TEN-C, FOU-D, FOU-H, QUE-D, NIN-S, TWO-D	FALSE
FIV-H, EIG-H, ACE-S, QUE-S, SIX-S, SIX-S	FALSE
EIG-S, JAC-S, TEN-S, NIN-S, JAC-D, KIN-H	TRUE
Holdout Examples (first 4):	
Hand	Label
FIV-S, SIX-C, NIN-S, ACE-D, SEV-D, TWO-C	FALSE
QUE-C, QUE-D, JAC-S, ACE-S, ACE-H, SEV-C	TRUE
NIN-S, ACE-S, THR-S, ACE-H, JAC-D, EIG-H	TRUE
QUE-H, NIN-H, NIN-H, JAC-S, SEV-C, NIN-S	FALSE

2.2.3 Ends_same_suit

Family: LOCAL, Difficulty: 4

Training Examples (first 6):

Hand	Label
FIV-D, TEN-S, THR-S, SEV-C, SEV-C, QUE-S	FALSE
TEN-C, TEN-D, SEV-H, QUE-D, TEN-S, EIG-D	FALSE
SEV-H, NIN-D, JAC-C, SIX-H, NIN-C, KIN-D	FALSE
FIV-H, QUE-D, JAC-C, SEV-S, FOU-H, KIN-H	TRUE
TEN-S, SEV-D, FOU-D, QUE-S, ACE-C, QUE-C	FALSE
EIG-H, NIN-D, FIV-S, FIV-S, THR-S, SIX-S	FALSE
Holdout Examples (first 4):	
Hand	Label
NIN-D, QUE-C, FIV-D, TEN-D, TEN-S, TEN-S	FALSE
FOU-S, NIN-S, FOU-S, TWO-S, THR-H, THR-S	TRUE
NIN-S, EIG-H, THR-C, NIN-S, FOU-C, TWO-C	FALSE
TEN-H, TWO-S, FIV-H, ACE-C, KIN-H, FOU-D	FALSE

2.2.4 Ends_same_color

Family: LOCAL, Difficulty: 4

Training Examples (first 6):

Hand	Label
TWO-D, FIV-H, TWO-C, KIN-C, QUE-H, KIN-H	TRUE
ACE-C, TWO-D, JAC-D, FOU-S, FOU-D, TEN-C	TRUE
ACE-S, THR-H, NIN-S, ACE-D, EIG-D, FIV-S	TRUE
THR-S, NIN-D, SIX-D, KIN-H, FIV-H, QUE-C	TRUE
EIG-C, JAC-S, SEV-S, SEV-D, TEN-S, FIV-S	TRUE
TWO-H, SIX-H, JAC-C, EIG-C, SIX-C, QUE-S	FALSE

Holdout Examples (first 4):

Hand	Label
QUE-D, NIN-H, FIV-S, SIX-C, JAC-D, JAC-S	FALSE
KIN-S, THR-C, TWO-H, FIV-H, FOU-H, ACE-S	TRUE
QUE-S, JAC-D, FIV-D, EIG-H, ACE-S, TWO-C	TRUE
SIX-D, EIG-S, TWO-H, QUE-C, QUE-S, EIG-H	TRUE

2.2.5 Has_pair_ranks

Family: COUNT, Difficulty: 1

Training Examples (first 6):

Hand	Label
SEV-D, TEN-S, SIX-D, ACE-S, NIN-S, EIG-S	FALSE
THR-S, THR-D, QUE-H, THR-C, JAC-H, FOU-C	TRUE
FOU-C, TWO-S, TEN-H, NIN-D, KIN-D, JAC-C	FALSE
TEN-C, TEN-S, FIV-C, QUE-H, EIG-D, EIG-H	TRUE
THR-C, SEV-C, ACE-D, JAC-C, FIV-S, FIV-C	TRUE
EIG-C, NIN-S, THR-S, KIN-C, TEN-H, SIX-H	FALSE

Holdout Examples (first 4):

Hand	Label
FOU-C, SIX-D, QUE-C, QUE-H, TEN-H, EIG-H	TRUE
FOU-D, SIX-D, EIG-S, SIX-S, ACE-H, KIN-C	TRUE
SEV-C, SIX-H, KIN-C, QUE-D, TWO-S, THR-S	FALSE
FIV-H, FIV-D, QUE-D, ACE-C, FIV-D, FIV-D	TRUE

2.2.6 Uniform_color

Family: COUNT, Difficulty: 1

Training Examples (first 6):

Hand	Label
THR-D, QUE-H, FIV-H, EIG-H, THR-D, FOU-H	TRUE
TEN-D, SEV-H, NIN-H, EIG-H, SIX-D, SIX-D	TRUE
JAC-C, NIN-H, NIN-C, TEN-C, NIN-C, TWO-S	FALSE
TWO-H, FIV-D, QUE-C, JAC-D, THR-D, SIX-D	FALSE
FIV-C, EIG-H, SIX-C, SIX-S, TEN-S, THR-S	FALSE
QUE-C, FOU-S, SIX-D, TWO-S, EIG-S, ACE-C	FALSE

Holdout Examples (first 4):

Hand	Label
NIN-D, JAC-D, JAC-S, SEV-S, FIV-S, TWO-D	FALSE
EIG-S, FOU-S, FOU-C, KIN-C, ACE-S, NIN-S	TRUE
FOU-D, KIN-H, JAC-H, FIV-D, EIG-H, FIV-H	TRUE
TWO-C, TWO-D, THR-S, FIV-S, TEN-S, FOU-S	FALSE

2.2.7 Exactly_two_suits

Family: COUNT, Difficulty: 1

Training Examples (first 6):

Hand	Label
SEV-H, TWO-D, KIN-S, EIG-D, SEV-D, EIG-H	FALSE
JAC-C, TEN-C, THR-C, QUE-H, JAC-C, SEV-C	TRUE
SIX-C, QUE-H, THR-H, SIX-C, FIV-C, THR-C	TRUE
FOU-H, ACE-C, SIX-H, SIX-C, SIX-S, FIV-H	FALSE
FOU-H, FIV-C, SIX-C, SEV-S, FIV-C, TEN-S	FALSE
JAC-C, QUE-H, FOU-D, FIV-C, EIG-H, FOU-H	FALSE

Holdout Examples (first 4):

Hand	Label
SIX-S, FIV-C, SIX-S, SIX-C, NIN-H, SEV-S	FALSE
TEN-C, SIX-S, TWO-C, SIX-S, SEV-S, QUE-C	TRUE
TWO-H, KIN-S, FOU-D, THR-C, TWO-S, FOU-H	FALSE
FIV-D, FOU-H, JAC-S, KIN-H, TWO-D, TEN-C	FALSE

2.2.8 Half_or_more_same_suit

Family: COUNT, Difficulty: 1

Training Examples (first 6):

Hand	Label
TWO-C, SEV-D, TWO-H, TEN-S, FIV-C, SIX-S	FALSE
FIV-D, ACE-S, FIV-S, JAC-S, SIX-H, FIV-D	TRUE
NIN-D, ACE-H, SIX-S, NIN-C, SIX-D, THR-S	FALSE
SIX-H, THR-D, SEV-S, FIV-D, SIX-D, EIG-C	TRUE
SEV-H, ACE-H, KIN-H, EIG-H, THR-S, FIV-H	TRUE
NIN-S, SEV-H, KIN-C, ACE-D, SEV-S, EIG-S	TRUE

Holdout Examples (first 4):

Hand	Label
EIG-H, FIV-S, TEN-H, TWO-H, ACE-S, TEN-S	TRUE
JAC-S, QUE-C, FOU-C, SEV-S, TEN-H, TEN-S	TRUE
NIN-S, QUE-C, ACE-C, TWO-H, FOU-H, THR-H	TRUE
NIN-D, KIN-S, THR-H, THR-D, FOU-S, QUE-C	FALSE

2.2.9 At_most_three_suits

Family: COUNT, Difficulty: 1

Training Examples (first 6):

Hand	Label
EIG-H, SEV-S, ACE-C, KIN-S, TEN-C, ACE-D	FALSE
ACE-H, NIN-H, ACE-D, FOU-H, FOU-H, TEN-C	TRUE
JAC-D, KIN-C, SEV-C, JAC-H, KIN-D, ACE-H	TRUE
ACE-H, NIN-H, ACE-D, SEV-S, TWO-D, KIN-H	TRUE
EIG-H, KIN-C, EIG-S, ACE-D, FOU-S, TEN-S	FALSE
JAC-H, TEN-H, TEN-S, ACE-C, FOU-S, TEN-D	FALSE

Holdout Examples (first 4):

Hand	Label
THR-D, KIN-C, THR-H, SIX-C, ACE-C, ACE-D	TRUE
EIG-H, FOU-S, NIN-D, FIV-C, EIG-S, THR-C	FALSE
TWO-H, THR-S, TWO-S, ACE-C, SEV-D, NIN-D	FALSE
THR-D, QUE-H, NIN-H, QUE-H, NIN-D, KIN-D	TRUE

2.2.10 Exactly_one_club

Family: COUNT, Difficulty: 1

Training Examples (first 6):

Hand	Label
NIN-C, TWO-D, SIX-S, TEN-S, NIN-S, JAC-S	TRUE
TWO-S, FOU-S, KIN-H, SIX-C, TEN-D, THR-D	TRUE
FOU-S, THR-H, ACE-S, EIG-D, JAC-S, ACE-D	FALSE
KIN-C, FOU-D, KIN-H, EIG-C, EIG-D, ACE-H	FALSE
THR-D, NIN-C, FIV-S, TEN-D, QUE-H, EIG-C	FALSE
EIG-D, SEV-H, EIG-C, FIV-C, THR-S, ACE-H	FALSE

Holdout Examples (first 4):

Hand	Label
FIV-D, ACE-S, EIG-C, SIX-S, JAC-S, QUE-D	TRUE
TWO-D, FIV-D, KIN-D, TEN-H, THR-H, FOU-C	TRUE
EIG-H, EIG-S, FIV-D, SEV-C, SEV-S, SIX-H	TRUE
KIN-C, THR-H, SEV-H, SIX-D, SIX-C, SEV-H	FALSE

2.2.11 AP_len3_anywhere_anyk

Family: AP, Difficulty: 3

Training Examples (first 6):

Hand	Label
TEN-S, TEN-D, FOU-H, QUE-C, QUE-H, ACE-C	TRUE
SEV-C, FIV-D, TWO-D, FIV-H, ACE-C, FOU-D	FALSE
TEN-S, TEN-H, TWO-S, SIX-H, EIG-H, JAC-C	TRUE
EIG-S, SIX-H, TWO-S, THR-D, SIX-D, SIX-S	TRUE
NIN-H, JAC-H, NIN-D, EIG-C, QUE-C, EIG-C	FALSE
NIN-H, EIG-S, KIN-D, SIX-H, ACE-C, FIV-D	TRUE

Holdout Examples (first 4):

Hand	Label
FIV-D, ACE-H, SIX-C, KIN-S, FIV-C, TWO-C	FALSE
THR-D, JAC-C, FIV-D, FIV-S, NIN-D, SEV-H	TRUE
FIV-S, TWO-D, SEV-D, KIN-D, FOU-D, FOU-H	FALSE
NIN-C, KIN-D, SIX-H, KIN-S, EIG-H, TWO-D	FALSE

2.2.12 AP_len3_step2_anywhere

Family: AP, Difficulty: 3

Training Examples (first 6):

Hand	Label
NIN-C, JAC-S, ACE-D, QUE-H, TEN-C, TEN-D	TRUE
ACE-D, NIN-H, FIV-D, NIN-C, TWO-S, SEV-C	TRUE
QUE-S, ACE-H, ACE-H, KIN-D, TEN-C, NIN-D	TRUE
TWO-S, EIG-S, SIX-C, EIG-H, SEV-H, QUE-S	FALSE
FOU-C, QUE-S, NIN-D, FIV-D, SEV-C, KIN-C	TRUE
FIV-H, NIN-C, QUE-S, TEN-C, NIN-C, SIX-H	FALSE

Holdout Examples (first 4):

Hand	Label
QUE-D, THR-D, SEV-H, FIV-H, ACE-C, EIG-D	TRUE
TEN-H, QUE-D, TWO-S, EIG-C, TWO-S, SIX-C	TRUE
SIX-S, QUE-C, ACE-S, TEN-C, THR-S, TWO-D	TRUE
SIX-C, KIN-C, JAC-H, EIG-C, ACE-D, SIX-H	FALSE

2.2.13 AP_len4_step2_anywhere

Family: AP, Difficulty: 3

Training Examples (first 6):

Hand	Label
THR-C, QUE-H, FIV-D, NIN-C, SEV-H, THR-C	TRUE
NIN-H, SIX-C, THR-S, JAC-D, FIV-H, SEV-C	TRUE
TEN-D, EIG-H, SIX-H, TWO-C, QUE-S, FOU-C	TRUE
EIG-D, SIX-C, JAC-S, FOU-D, TWO-H, NIN-D	TRUE
JAC-S, THR-H, FIV-H, FIV-C, SEV-C, NIN-H	TRUE
TWO-H, QUE-C, EIG-H, KIN-C, QUE-H, TEN-C	FALSE

Holdout Examples (first 4):

Hand	Label
ACE-C, THR-C, FIV-C, SEV-S, SIX-C, SIX-H	FALSE
JAC-H, KIN-H, KIN-S, KIN-C, SIX-C, QUE-S	FALSE
SIX-S, SEV-C, TEN-H, FOU-H, TWO-H, EIG-S	TRUE
QUE-D, SIX-S, FIV-D, EIG-H, ACE-C, ACE-C	FALSE

2.2.14 Half_sum_diff_geN

Family: SCORE, Difficulty: 2

Training Examples (first 6):

Hand	Label
KIN-H, SEV-D, SEV-H, FOU-S, ACE-S, FIV-H	FALSE
EIG-S, THR-C, TEN-H, NIN-H, FIV-D, THR-C	FALSE
QUE-D, NIN-S, THR-S, FOU-S, JAC-D, TWO-D	TRUE
JAC-S, JAC-H, KIN-C, FIV-D, ACE-D, JAC-S	FALSE
THR-D, THR-C, NIN-D, THR-H, THR-D, TWO-H	TRUE
EIG-H, SIX-D, FIV-S, FIV-H, TWO-S, SIX-H	TRUE

Holdout Examples (first 4):

Hand	Label
JAC-C, ACE-D, THR-H, FOU-H, FIV-H, QUE-H	TRUE
KIN-C, ACE-H, THR-D, SIX-S, NIN-H, THR-H	TRUE
NIN-D, ACE-S, FIV-C, SIX-H, NIN-D, FOU-S	TRUE
EIG-S, ACE-S, THR-D, TWO-S, FIV-S, TEN-H	TRUE

2.2.15 Half_sum_one_side_ge_2x_other

Family: SCORE, Difficulty: 2

Training Examples (first 6):

Hand	Label
QUE-D, QUE-H, EIG-H, SIX-S, ACE-D, THR-D	FALSE
TWO-S, TWO-D, FIV-C, SEV-D, FOU-D, JAC-C	TRUE
TEN-S, JAC-D, JAC-S, NIN-H, THR-H, THR-H	TRUE
SIX-C, QUE-D, KIN-S, NIN-S, TWO-S, KIN-C	FALSE
JAC-C, TWO-H, TEN-C, FOU-H, FIV-H, FOU-C	FALSE
FIV-H, THR-D, FIV-H, JAC-H, QUE-S, JAC-D	TRUE

Holdout Examples (first 4):

Hand	Label
SIX-C, FIV-D, THR-D, KIN-S, ACE-S, TEN-S	TRUE
THR-S, FOU-D, FOU-S, SIX-D, ACE-H, KIN-D	TRUE
NIN-S, TWO-H, THR-H, FOU-D, KIN-S, KIN-S	TRUE
THR-D, EIG-H, TWO-S, SIX-C, QUE-H, JAC-D	TRUE

2.2.16 Halves_uniform_color_equal

Family: HIER, Difficulty: 2

Training Examples (first 6):

Hand	Label
TWO-S, FIV-S, JAC-S, JAC-D, QUE-C, NIN-S	FALSE
SEV-C, SEV-S, TWO-S, TEN-H, FIV-S, NIN-C	FALSE
TEN-C, THR-D, ACE-S, SEV-C, FOU-C, KIN-C	FALSE
NIN-D, EIG-H, FIV-S, SIX-H, TWO-C, FOU-S	TRUE
FOU-D, EIG-C, FIV-C, SEV-C, TEN-S, FIV-C	FALSE
NIN-C, EIG-D, SIX-D, TWO-H, TEN-S, QUE-H	TRUE

Holdout Examples (first 4):

Hand	Label
TEN-H, FIV-S, TEN-H, NIN-S, TWO-H, SEV-S	TRUE
EIG-C, KIN-C, SIX-S, TEN-H, ACE-D, QUE-S	FALSE
THR-C, TEN-D, JAC-C, NIN-H, TEN-H, JAC-D	FALSE
SIX-D, ACE-D, TWO-H, SIX-D, THR-C, JAC-S	FALSE

2.2.17 Halves_AP_step1_equal

Family: HIER, Difficulty: 2

Training Examples (first 6):

Hand	Label
NIN-C, FIV-H, KIN-H, QUE-D, FIV-C, EIG-S	TRUE
EIG-H, SIX-D, JAC-H, TEN-C, JAC-S, QUE-D	FALSE
FIV-C, KIN-C, ACE-S, EIG-S, TWO-S, TWO-S	TRUE
JAC-D, ACE-S, QUE-H, FOU-C, NIN-C, THR-S	TRUE
QUE-S, SIX-S, TEN-H, ACE-S, JAC-D, THR-S	TRUE
SIX-H, THR-C, ACE-D, FOU-C, SEV-C, QUE-S	TRUE

Holdout Examples (first 4):

Hand	Label
JAC-D, NIN-C, SEV-S, FIV-S, FOU-H, JAC-S	TRUE
QUE-C, THR-D, FOU-D, TEN-H, TWO-S, TEN-D	TRUE
SIX-C, ACE-H, TEN-D, THR-H, QUE-H, NIN-D	TRUE
FOU-D, FIV-C, SIX-S, ACE-H, KIN-H, FIV-S	FALSE

2.2.18 Halves_hearts_presence_equal

Family: HIER, Difficulty: 2

Training Examples (first 6):

Hand	Label
EIG-H, FOU-D, THR-S, EIG-H, EIG-C, ACE-S	TRUE
TEN-H, TWO-D, ACE-S, THR-D, EIG-D, FOU-D	FALSE
SIX-H, QUE-H, EIG-D, SEV-S, JAC-D, TWO-H	TRUE
SIX-S, SEV-H, SEV-D, TWO-C, SIX-H, JAC-S	TRUE
JAC-C, TEN-C, FIV-D, TEN-D, FIV-C, SEV-D	TRUE
JAC-S, QUE-H, SIX-H, ACE-C, SEV-H, SEV-H	TRUE

Holdout Examples (first 4):

Hand	Label
TEN-H, QUE-C, FIV-H, TEN-S, QUE-S, FIV-D	FALSE
FIV-H, JAC-D, SIX-C, THR-C, SEV-C, TEN-H	TRUE
ACE-C, FIV-D, TWO-C, KIN-S, THR-C, THR-D	TRUE
ACE-C, NIN-S, EIG-S, QUE-H, THR-S, FOU-S	FALSE

2.2.19 Halves_AP_len3_any_equal

Family: HIER, Difficulty: 3

Training Examples (first 6):

Hand	Label
SIX-S, FOU-S, ACE-H, FIV-D, KIN-S, FOU-C	TRUE
FIV-C, ACE-D, EIG-H, FOU-H, EIG-D, KIN-C	TRUE
FIV-D, SIX-D, SEV-D, JAC-H, TWO-S, FOU-H	FALSE
KIN-C, TWO-S, SEV-C, QUE-C, QUE-C, QUE-S	FALSE
SEV-C, FIV-C, FIV-D, SIX-C, NIN-S, QUE-C	FALSE
FOU-D, SEV-C, THR-D, EIG-S, SEV-D, NIN-D	FALSE

Holdout Examples (first 4):

Hand	Label
SEV-S, SIX-D, NIN-S, TEN-C, QUE-C, QUE-C	TRUE
SIX-C, NIN-H, JAC-C, TWO-D, SIX-S, JAC-D	TRUE
KIN-C, TEN-C, THR-S, FIV-S, TEN-H, SIX-D	TRUE
NIN-D, NIN-S, NIN-H, TWO-C, FIV-D, TEN-D	FALSE

2.2.20 Halves_AP_len2_step1_equal

Family: HIER, Difficulty: 2

Training Examples (first 6):

Hand	Label
SEV-D, NIN-C, QUE-H, SIX-S, JAC-S, FOU-S	TRUE
QUE-C, EIG-C, SIX-S, QUE-D, KIN-H, TEN-C	FALSE
KIN-H, FOU-S, NIN-S, NIN-H, KIN-S, EIG-H	FALSE
FOU-C, FIV-C, EIG-H, ACE-C, KIN-D, TEN-H	TRUE
TWO-H, THR-S, TEN-C, KIN-S, THR-D, JAC-S	FALSE
TWO-H, NIN-D, FOU-S, FOU-D, TEN-S, SEV-H	TRUE

Holdout Examples (first 4):

Hand	Label
SEV-C, NIN-D, TWO-S, SEV-H, JAC-H, SEV-D	TRUE
EIG-S, THR-S, EIG-C, ACE-C, EIG-S, THR-D	TRUE
SIX-D, FIV-H, NIN-S, QUE-D, NIN-H, JAC-H	TRUE
FOU-H, SIX-C, KIN-C, ACE-S, TWO-S, THR-S	FALSE

2.2.21 Well_formed_brackets_by_suit

Family: LANG, Difficulty: 3

Training Examples (first 6):

Hand	Label
SIX-H, NIN-S, EIG-D, FIV-H, ACE-C, QUE-D	FALSE
ACE-H, NIN-C, EIG-D, QUE-C, SIX-D, THR-H	FALSE
TWO-D, NIN-D, SIX-D, THR-S, ACE-C, NIN-H	TRUE
KIN-D, KIN-C, EIG-D, TEN-S, KIN-C, SIX-D	FALSE
SIX-C, ACE-H, TEN-S, QUE-C, SIX-H, QUE-D	FALSE
EIG-S, KIN-C, SEV-C, QUE-H, NIN-C, TWO-C	TRUE

Holdout Examples (first 4):

Hand	Label
FOU-H, THR-D, SEV-D, NIN-S, NIN-H, KIN-C	FALSE
FIV-C, NIN-C, ACE-C, QUE-H, NIN-S, SEV-C	FALSE
FOU-C, ACE-H, SEV-S, JAC-H, QUE-H, JAC-D	FALSE
KIN-D, TWO-C, EIG-S, FOU-H, FOU-C, ACE-C	TRUE

2.2.22 Even_opens_next_closes

Family: LANG, Difficulty: 3

Training Examples (first 6):

Hand	Label
TEN-D, TEN-S, THR-D, KIN-S, THR-S, FIV-H	FALSE
EIG-C, SEV-S, QUE-H, FOU-H, KIN-D, QUE-S	FALSE
FOU-C, FOU-D, SEV-S, THR-D, ACE-C, SIX-D	FALSE
EIG-S, EIG-C, JAC-D, TEN-D, JAC-D, FIV-D	TRUE
JAC-S, SIX-C, FIV-C, ACE-C, NIN-S, FIV-C	FALSE
ACE-D, NIN-C, EIG-S, FIV-D, TEN-S, SEV-S	TRUE

Holdout Examples (first 4):

Hand	Label
SEV-S, ACE-C, EIG-S, EIG-D, QUE-S, ACE-S	FALSE
TWO-D, JAC-H, THR-H, QUE-C, FOU-H, EIG-H	FALSE
TEN-S, THR-H, TEN-C, TEN-C, THR-H, SEV-C	TRUE
SEV-S, THR-C, JAC-D, FIV-C, FIV-D, FOU-H	FALSE

2.2.23 Odd_opens_next_closes

Family: LANG, Difficulty: 3

Training Examples (first 6):

Hand	Label
KIN-C, KIN-S, SEV-S, NIN-C, FOU-D, SIX-S	FALSE
JAC-D, EIG-S, NIN-C, NIN-C, FOU-S, TWO-S	TRUE
FIV-D, JAC-S, ACE-S, SEV-C, QUE-H, FOU-C	TRUE
FIV-D, KIN-D, NIN-S, THR-H, SIX-S, EIG-S	FALSE
SEV-C, JAC-C, TEN-S, TWO-H, NIN-H, TWO-H	TRUE
THR-C, SEV-S, FOU-S, TEN-D, ACE-H, ACE-D	FALSE

Holdout Examples (first 4):

Hand	Label
THR-S, JAC-C, ACE-H, TWO-D, KIN-S, TWO-H	TRUE
JAC-D, FOU-D, KIN-S, THR-D, FOU-C, TWO-S	TRUE
FIV-D, SIX-C, ACE-H, JAC-S, FIV-S, JAC-H	FALSE
JAC-D, TEN-S, JAC-C, FOU-D, KIN-C, TWO-C	TRUE

2.2.24 Suits_palindrome

Family: PAL, Difficulty: 4

Training Examples (first 6):

Hand	Label
TEN-D, EIG-S, SEV-D, QUE-D, TEN-S, FIV-D	TRUE
JAC-D, TEN-D, EIG-S, QUE-S, JAC-D, FIV-D	TRUE
TEN-D, JAC-C, JAC-H, QUE-H, TEN-C, SEV-S	FALSE
FIV-H, KIN-H, SIX-C, TEN-C, FOU-H, THR-C	FALSE
JAC-D, ACE-S, SIX-D, TWO-H, FOU-S, QUE-D	FALSE
EIG-H, KIN-H, NIN-C, QUE-C, NIN-H, JAC-H	TRUE

Holdout Examples (first 4):

Hand	Label
TWO-C, SIX-H, FOU-C, SIX-C, SEV-H, ACE-C	TRUE
SEV-S, QUE-S, NIN-H, FOU-D, EIG-S, QUE-H	FALSE
TEN-S, SIX-D, SEV-D, EIG-D, FOU-H, FIV-D	FALSE
NIN-D, FOU-S, TEN-S, SEV-S, FOU-S, SEV-D	TRUE

2.2.25 Colors_palindrome

Family: PAL, Difficulty: 4

Training Examples (first 6):

Hand	Label
KIN-H, TEN-C, SEV-C, THR-S, TEN-S, THR-D	TRUE
SIX-D, KIN-D, FIV-S, TWO-S, NIN-D, FIV-H	TRUE
FOU-H, TEN-C, ACE-S, QUE-C, TEN-C, JAC-C	FALSE
EIG-S, SEV-C, NIN-H, SEV-H, THR-H, SEV-S	FALSE
NIN-D, QUE-H, FOU-C, SEV-S, THR-H, FOU-H	TRUE
EIG-H, FIV-D, FIV-S, TWO-S, NIN-D, KIN-H	TRUE

Holdout Examples (first 4):

Hand	Label
EIG-H, KIN-D, THR-C, SEV-C, FOU-C, NIN-C	FALSE
FIV-D, TWO-S, TWO-D, ACE-H, THR-S, NIN-D	TRUE
EIG-D, JAC-H, QUE-C, KIN-S, QUE-C, JAC-H	FALSE
ACE-H, TEN-D, EIG-C, THR-D, FOU-C, QUE-S	FALSE

2.2.26 Ranks_palindrome

Family: PAL, Difficulty: 4

Training Examples (first 6):

Hand	Label
THR-H, SIX-S, FOU-S, FOU-H, SIX-C, THR-D	TRUE
SIX-C, FIV-D, ACE-D, ACE-S, FIV-C, TEN-D	FALSE
TEN-H, ACE-C, FIV-C, TEN-D, ACE-H, TEN-D	FALSE
JAC-D, SIX-C, ACE-C, ACE-S, SIX-D, JAC-H	TRUE
KIN-D, FIV-D, SIX-D, SIX-S, FIV-S, KIN-C	TRUE
TWO-C, JAC-H, EIG-H, ACE-S, JAC-S, TWO-S	FALSE

Holdout Examples (first 4):

Hand	Label
TEN-C, TWO-H, FOU-H, FOU-C, TWO-S, TEN-H	TRUE
JAC-H, FOU-H, THR-H, THR-H, FOU-C, JAC-D	TRUE
JAC-S, ACE-C, THR-C, THR-C, ACE-D, JAC-H	TRUE
FOU-S, FOU-D, FOU-H, TEN-H, EIG-S, KIN-C	FALSE

2.2.27 AltColor1_palindrome

Family: ALTCLR, Difficulty: 4

Training Examples (first 6):

Hand	Label
EIG-S, EIG-C, TEN-S, NIN-H, SIX-H, SIX-D	FALSE
ACE-S, NIN-S, TEN-D, FIV-S, QUE-H, FOU-D	FALSE
TWO-H, TWO-S, FIV-H, KIN-H, TWO-D, JAC-H	TRUE
FOU-H, TWO-H, SEV-C, FOU-H, JAC-H, QUE-C	TRUE
TEN-C, THR-H, QUE-D, EIG-S, TWO-C, TWO-S	FALSE
THR-H, SIX-D, FOU-C, ACE-C, SEV-S, KIN-H	TRUE

Holdout Examples (first 4):

Hand	Label
TWO-C, TEN-H, EIG-D, TWO-D, KIN-C, THR-D	FALSE
SIX-C, FOU-D, TWO-H, TEN-H, QUE-S, ACE-C	TRUE
SEV-C, TWO-C, FIV-H, KIN-H, KIN-C, SEV-C	TRUE
FIV-C, TWO-H, SIX-S, TEN-D, FIV-C, JAC-H	TRUE

2.2.28 AltColor2_palindrome**Family:** ALTCLR, **Difficulty:** 4**Training Examples (first 6):**

Hand	Label
ACE-C, ACE-C, JAC-S, NIN-D, EIG-C, TWO-C	FALSE
SIX-C, KIN-S, KIN-S, JAC-H, QUE-D, SIX-C	FALSE
SIX-C, KIN-C, TWO-D, ACE-S, THR-C, THR-D	FALSE
TEN-C, SIX-H, QUE-H, KIN-H, THR-H, TWO-S	FALSE
TEN-C, EIG-D, SIX-D, THR-S, FOU-D, EIG-D	FALSE
JAC-S, TWO-D, QUE-D, JAC-D, NIN-D, TEN-H	TRUE

Holdout Examples (first 4):

Hand	Label
TWO-C, EIG-S, ACE-S, NIN-S, FIV-H, SIX-C	TRUE
ACE-C, THR-S, KIN-D, FOU-D, TEN-S, QUE-H	FALSE
SEV-H, TEN-H, TEN-S, SEV-H, NIN-H, QUE-S	TRUE
FOU-S, ACE-H, NIN-H, TEN-S, ACE-S, KIN-S	TRUE

2.2.29 Ends_same_altcolor1**Family:** ALTCLR, **Difficulty:** 4**Training Examples (first 6):**

Hand	Label
JAC-H, TEN-H, JAC-H, QUE-D, FOU-S, SIX-S	FALSE
TWO-H, TEN-C, FIV-D, QUE-D, JAC-D, NIN-H	TRUE
FOU-C, SEV-H, NIN-H, QUE-C, ACE-C, THR-D	FALSE
QUE-C, FOU-C, EIG-D, THR-H, NIN-H, JAC-S	FALSE
SIX-D, TEN-S, ACE-C, SIX-D, FOU-C, TEN-D	TRUE
EIG-H, TWO-C, FIV-D, THR-C, QUE-H, SEV-D	FALSE

Holdout Examples (first 4):

Hand	Label
FIV-D, EIG-C, NIN-H, FOU-D, TEN-H, KIN-D	TRUE
NIN-D, SIX-C, FOU-S, SIX-C, NIN-H, TEN-S	TRUE
KIN-H, JAC-D, FIV-H, THR-D, FOU-S, FIV-H	TRUE
TWO-D, TWO-H, FIV-C, ACE-H, SIX-H, SEV-S	TRUE

2.2.30 Halves_copy_suits**Family:** COPY, **Difficulty:** 4**Training Examples (first 6):**

Hand	Label
NIN-H, NIN-D, SIX-S, JAC-H, TEN-D, NIN-H	FALSE
SEV-S, KIN-H, QUE-C, NIN-S, JAC-H, JAC-C	TRUE
SIX-D, QUE-S, TWO-C, THR-D, TEN-S, FOU-C	TRUE
THR-H, FOU-S, QUE-H, SEV-H, NIN-C, QUE-H	FALSE
QUE-H, ACE-H, SIX-S, QUE-H, NIN-H, QUE-C	FALSE
TEN-C, ACE-D, ACE-H, FOU-C, NIN-D, ACE-H	TRUE

Holdout Examples (first 4):

Hand	Label
QUE-H, ACE-S, FOU-D, THR-S, QUE-D, ACE-S	FALSE
FIV-S, THR-C, SIX-C, EIG-S, ACE-C, TEN-C	TRUE
NIN-H, SEV-S, EIG-S, TEN-S, FIV-H, EIG-C	FALSE
THR-D, SEV-D, TWO-C, KIN-D, FOU-D, KIN-C	TRUE

2.2.31 Halves_copy_colors

Family: COPY, Difficulty: 4

Training Examples (first 6):

Hand	Label
QUE-S, TWO-C, ACE-C, ACE-S, FIV-S, ACE-S	TRUE
FIV-C, KIN-S, TEN-H, TEN-S, THR-D, TWO-H	FALSE
FIV-D, KIN-H, FIV-C, JAC-D, JAC-H, SIX-C	TRUE
EIG-C, NIN-D, ACE-S, FIV-C, TWO-D, THR-D	FALSE
KIN-C, FIV-D, FIV-S, QUE-C, FIV-S, NIN-C	FALSE
FIV-H, FIV-H, QUE-S, KIN-D, THR-H, NIN-C	TRUE

Holdout Examples (first 4):

Hand	Label
NIN-D, FOU-C, SEV-S, SEV-D, JAC-S, THR-C	TRUE
EIG-S, KIN-S, QUE-C, SIX-S, FIV-C, SIX-C	TRUE
TWO-C, KIN-C, SIX-S, SEV-D, KIN-D, FOU-S	FALSE
SEV-D, QUE-H, TEN-S, FIV-H, SEV-C, TWO-C	FALSE

2.2.32 Halves_copy_ranks

Family: COPY, Difficulty: 4

Training Examples (first 6):

Hand	Label
JAC-H, SIX-S, TEN-D, JAC-S, SIX-H, TEN-S	TRUE
TWO-S, TWO-D, EIG-S, TWO-D, THR-H, EIG-D	FALSE
TEN-H, EIG-H, SIX-S, TEN-C, EIG-D, THR-C	FALSE
QUE-H, ACE-C, TEN-H, QUE-C, ACE-C, TEN-D	TRUE
JAC-C, TEN-C, FIV-S, JAC-S, TEN-C, FIV-S	TRUE
QUE-H, QUE-H, NIN-S, QUE-S, QUE-C, QUE-S	FALSE

Holdout Examples (first 4):

Hand	Label
NIN-D, KIN-H, FIV-C, NIN-H, QUE-D, KIN-H	FALSE
SIX-D, FOU-C, QUE-D, EIG-D, FOU-H, NIN-D	FALSE
QUE-C, TEN-D, FIV-C, TWO-C, JAC-C, THR-D	FALSE
NIN-H, SEV-H, QUE-C, NIN-C, SEV-C, QUE-H	TRUE

2.2.33 Halves_copy_altcolor1

Family: COPY, Difficulty: 4

Training Examples (first 6):

Hand	Label
NIN-H, KIN-C, KIN-S, QUE-C, FIV-C, ACE-D	TRUE
SEV-H, SEV-D, TWO-S, TEN-C, TWO-H, JAC-D	FALSE
SEV-H, SIX-H, FIV-D, TWO-C, SEV-D, TEN-D	FALSE
FIV-D, FOU-S, SEV-S, FOU-S, THR-D, FOU-S	TRUE
EIG-H, ACE-D, QUE-C, THR-C, SIX-D, NIN-H	TRUE
EIG-S, THR-S, NIN-D, EIG-S, SIX-D, FOU-D	TRUE

Holdout Examples (first 4):

Hand	Label
QUE-C, TEN-D, NIN-S, KIN-D, SEV-H, QUE-C	FALSE
JAC-C, TWO-D, TEN-C, JAC-C, JAC-S, ACE-C	TRUE
ACE-S, SIX-C, KIN-H, EIG-S, FOU-H, JAC-C	TRUE
THR-D, FOU-C, SIX-C, THR-D, TEN-H, TEN-D	FALSE

2.2.34 Halves_copy_altcolor2

Family: COPY, Difficulty: 4

Training Examples (first 6):

Hand	Label
FOU-C, TWO-D, TEN-D, THR-D, TEN-D, JAC-H	FALSE
KIN-C, NIN-C, THR-D, EIG-D, FOU-C, FOU-D	TRUE
THR-D, TEN-H, QUE-D, TWO-C, NIN-H, SIX-C	TRUE
SIX-D, TEN-C, TEN-D, TEN-D, SEV-D, SEV-H	FALSE
ACE-H, THR-D, SEV-H, SEV-H, TEN-D, NIN-H	TRUE
TEN-H, JAC-H, FOU-D, THR-S, THR-S, TWO-H	FALSE

Holdout Examples (first 4):

Hand	Label
NIN-D, FOU-D, ACE-H, TEN-D, TWO-C, SIX-H	TRUE
EIG-H, QUE-S, ACE-C, FIV-H, SIX-S, SEV-D	TRUE
FIV-S, NIN-D, SIX-S, FIV-C, ACE-C, TWO-S	FALSE
THR-C, SIX-H, FOU-S, TEN-D, EIG-H, KIN-C	FALSE

2.2.35 Halves_same_suit_set

Family: COPY, Difficulty: 2

Training Examples (first 6):

Hand	Label
THR-C, FOU-C, FOU-S, KIN-C, TWO-C, FOU-C	FALSE
FOU-C, FIV-H, JAC-D, QUE-C, THR-D, FOU-D	FALSE
THR-D, JAC-S, JAC-D, NIN-D, SEV-D, TWO-S	TRUE
THR-S, THR-C, JAC-S, TEN-S, ACE-C, THR-S	TRUE
JAC-D, KIN-H, ACE-C, TEN-H, QUE-D, FIV-C	TRUE
TWO-H, QUE-C, ACE-H, TEN-C, SIX-H, TWO-C	TRUE

Holdout Examples (first 4):

Hand	Label
QUE-D, ACE-D, SEV-D, NIN-S, SIX-C, KIN-S	FALSE
FIV-C, FIV-S, ACE-H, JAC-D, SEV-H, SIX-H	FALSE
NIN-S, FOU-H, ACE-D, SEV-C, ACE-C, ACE-S	FALSE
KIN-C, SEV-S, SIX-D, KIN-S, FIV-D, SEV-C	TRUE

2.2.36 Shift_half_ge

Family: SHIFT, Difficulty: 2

Training Examples (first 6):

Hand	Label
FIV-H, QUE-H, THR-S, KIN-H, JAC-H, JAC-H	FALSE
NIN-H, QUE-H, FIV-C, JAC-C, FIV-D, JAC-D	FALSE
TWO-H, NIN-C, NIN-S, NIN-D, QUE-D, KIN-S	TRUE
FIV-S, ACE-S, TEN-S, QUE-H, TWO-C, JAC-S	FALSE
SIX-S, EIG-H, KIN-H, SEV-S, NIN-H, NIN-C	FALSE
QUE-C, ACE-C, SIX-C, KIN-C, ACE-D, SIX-C	TRUE

Holdout Examples (first 4):

Hand	Label
NIN-D, EIG-H, EIG-D, ACE-D, EIG-D, KIN-C	TRUE
QUE-H, FIV-C, TEN-S, TEN-H, EIG-D, THR-C	FALSE
FOU-D, TWO-D, THR-C, EIG-D, QUE-D, SEV-D	TRUE
NIN-S, JAC-S, EIG-D, TWO-D, JAC-D, TEN-D	FALSE

2.2.37 Adj_same_rank_or_suit

Family: ADJ, Difficulty: 2

Training Examples (first 6):

Hand	Label
FOU-S, THR-S, THR-S, THR-S, SEV-S, SEV-C	TRUE
QUE-D, ACE-D, ACE-H, JAC-H, ACE-H, QUE-H	TRUE
FOU-H, FIV-C, JAC-C, QUE-C, QUE-C, QUE-S	FALSE
SEV-C, TEN-D, SIX-D, TWO-D, TWO-H, TEN-H	FALSE
ACE-H, FOU-H, TWO-H, ACE-H, SEV-H, SEV-C	TRUE
JAC-D, EIG-C, FIV-D, SIX-D, JAC-D, ACE-D	FALSE

Holdout Examples (first 4):

Hand	Label
QUE-S, KIN-S, JAC-D, SEV-S, SIX-D, SIX-H	FALSE
QUE-C, THR-S, FIV-D, THR-S, THR-S, JAC-D	FALSE
TEN-S, KIN-S, JAC-S, JAC-C, ACE-C, JAC-C	TRUE
SEV-S, SEV-C, QUE-C, QUE-C, THR-C, FOU-C	TRUE

2.2.38 Skip2_same_rank_or_suit

Family: ADJ, Difficulty: 2

Training Examples (first 6):

Hand	Label
NIN-H, JAC-C, NIN-S, NIN-C, NIN-D, TEN-C	TRUE
FOU-C, NIN-C, FIV-C, QUE-C, NIN-S, SIX-C	FALSE
EIG-H, NIN-D, TWO-H, NIN-H, SEV-C, SIX-H	FALSE
NIN-H, KIN-S, THR-H, KIN-H, TEN-H, SEV-S	FALSE
FOU-S, SIX-C, FOU-C, NIN-S, NIN-C, KIN-S	FALSE
NIN-C, TWO-D, NIN-D, TEN-D, TEN-D, EIG-D	TRUE

Holdout Examples (first 4):

Hand	Label
TWO-D, KIN-D, JAC-C, QUE-D, QUE-S, ACE-C	FALSE
SEV-H, FIV-S, TEN-D, JAC-H, JAC-D, KIN-C	FALSE
SEV-H, TEN-S, SIX-S, ACE-D, THR-D, FOU-C	FALSE
JAC-S, SEV-S, TEN-D, FIV-H, THR-H, EIG-H	FALSE

2.2.39 Adj_rank_gap_le3

Family: ADJ, Difficulty: 2

Training Examples (first 6):

Hand	Label
JAC-C, TWO-C, JAC-C, TEN-D, QUE-S, TEN-C	FALSE
TEN-H, JAC-H, TWO-S, JAC-H, KIN-S, ACE-H	FALSE
TWO-C, FOU-C, SIX-S, QUE-H, FOU-D, FOU-C	FALSE
NIN-D, NIN-D, SIX-D, FIV-C, FIV-D, FIV-C	TRUE
THR-D, SIX-S, QUE-D, FOU-D, SIX-C, EIG-S	FALSE
THR-S, SIX-C, SIX-H, NIN-H, TEN-C, QUE-H	TRUE

Holdout Examples (first 4):

Hand	Label
JAC-C, JAC-C, NIN-H, JAC-C, NIN-D, KIN-D	FALSE
FOU-C, FOU-C, FIV-H, FIV-D, TEN-S, TEN-C	FALSE
JAC-D, EIG-S, QUE-D, KIN-H, FOU-C, KIN-S	FALSE
FIV-H, TEN-H, FIV-H, FOU-C, KIN-S, KIN-D	FALSE

2.2.40 Adj_same_rank_or_color

Family: ADJ, Difficulty: 2

Training Examples (first 6):

Hand	Label
TWO-S, NIN-C, JAC-C, QUE-S, EIG-H, JAC-S	FALSE
NIN-D, EIG-C, KIN-H, NIN-H, FIV-H, EIG-D	FALSE
ACE-D, FOU-D, SIX-D, THR-H, SIX-C, SEV-D	FALSE
ACE-S, JAC-H, QUE-S, FOU-S, THR-C, TEN-S	FALSE
JAC-C, SIX-S, TWO-C, KIN-S, NIN-S, THR-S	TRUE
THR-S, THR-D, JAC-D, TEN-H, FOU-D, NIN-C	FALSE

Holdout Examples (first 4):

Hand	Label
SEV-C, TWO-S, TWO-C, KIN-S, KIN-H, TWO-H	TRUE
QUE-D, KIN-C, SIX-H, FOU-S, THR-H, SIX-C	FALSE
FOU-H, FIV-D, KIN-D, FIV-D, SEV-S, TEN-C	FALSE
KIN-C, SIX-H, FIV-S, FOU-C, TWO-S, SIX-D	FALSE

2.2.41 Only_one_odd_rank

Family: PARITY, Difficulty: 1

Training Examples (first 6):

Hand	Label
EIG-S, TEN-H, KIN-C, ACE-C, THR-D, QUE-D	FALSE
FOU-C, TWO-C, ACE-C, FOU-D, SEV-C, ACE-H	TRUE
TEN-S, FIV-S, ACE-C, ACE-S, FIV-H, FOU-D	FALSE
SIX-H, TWO-H, SEV-S, JAC-H, FOU-S, QUE-H	FALSE
TWO-H, FIV-D, TWO-D, SIX-C, QUE-C, EIG-H	TRUE
EIG-H, FOU-D, TEN-S, SEV-C, TEN-H, EIG-D	TRUE

Holdout Examples (first 4):

Hand	Label
FIV-D, QUE-H, NIN-H, SIX-H, JAC-H, KIN-S	FALSE
TEN-D, QUE-D, NIN-D, SIX-S, FIV-S, FIV-H	FALSE
FOU-H, EIG-C, QUE-S, QUE-D, TEN-H, NIN-S	TRUE
QUE-S, FOU-C, FOU-D, FIV-D, TEN-S, ACE-H	TRUE

2.2.42 Uniform_rank_parity

Family: PARITY, Difficulty: 1

Training Examples (first 6):

Hand	Label
ACE-H, ACE-S, SIX-S, TEN-C, KIN-S, ACE-S	FALSE
SIX-S, TWO-S, FOU-H, KIN-D, ACE-C, EIG-S	FALSE
ACE-C, TWO-S, TWO-H, EIG-D, ACE-H, QUE-D	TRUE
EIG-H, QUE-H, SIX-S, NIN-D, SIX-D, ACE-C	FALSE
TEN-D, FOU-C, TWO-S, TWO-S, ACE-S, QUE-S	TRUE
TWO-C, TWO-H, FOU-H, EIG-S, QUE-H, TEN-C	TRUE

Holdout Examples (first 4):

Hand	Label
TEN-S, SEV-C, EIG-D, THR-S, NIN-C, KIN-D	FALSE
ACE-D, QUE-H, TWO-D, EIG-H, SIX-H, FOU-C	TRUE
QUE-S, EIG-D, TWO-C, JAC-S, SEV-C, THR-H	FALSE
FIV-D, SEV-D, SEV-C, JAC-H, KIN-C, FIV-H	TRUE

2.2.43 Halves_radial_nonincreasing**Family:** CENTER, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
NIN-C, QUE-D, QUE-D, QUE-H, SEV-D, NIN-C	FALSE
THR-D, TEN-S, QUE-H, FOU-S, FOU-S, FOU-S	TRUE
TEN-H, FIV-H, TEN-S, KIN-D, TEN-D, EIG-H	FALSE
THR-H, TWO-S, ACE-S, ACE-C, QUE-S, THR-D	FALSE
THR-D, FIV-C, SIX-D, TEN-S, NIN-H, FOU-C	TRUE
SEV-D, QUE-S, TEN-S, JAC-D, JAC-S, FIV-H	FALSE

Holdout Examples (first 4):

Hand	Label
FIV-S, THR-S, SIX-D, SIX-C, FIV-C, TEN-C	FALSE
SIX-H, NIN-D, TEN-H, THR-C, JAC-D, THR-C	FALSE
QUE-D, QUE-D, KIN-D, QUE-H, JAC-S, FOU-C	TRUE
QUE-D, TWO-D, QUE-C, NIN-D, TEN-D, NIN-S	FALSE

2.2.44 Global_radial_no_dominance**Family:** CENTER, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
FOU-D, TWO-H, QUE-S, KIN-H, FIV-D, THR-S	FALSE
TWO-D, SIX-H, EIG-H, NIN-H, KIN-S, FOU-H	FALSE
SIX-S, TEN-H, KIN-H, QUE-C, JAC-S, THR-S	TRUE
FOU-D, NIN-D, TEN-D, NIN-D, NIN-H, FOU-C	TRUE
SEV-S, SEV-D, ACE-S, ACE-H, ACE-H, ACE-S	FALSE
ACE-H, TEN-C, KIN-D, ACE-D, JAC-D, NIN-S	FALSE

Holdout Examples (first 4):

Hand	Label
SIX-D, THR-H, SEV-C, FIV-H, SIX-C, FIV-S	FALSE
FOU-C, TEN-S, QUE-S, JAC-S, TEN-H, SEV-D	TRUE
THR-D, KIN-D, ACE-S, FIV-C, FOU-H, SEV-H	FALSE
TWO-D, SIX-C, KIN-C, SEV-D, THR-H, THR-H	TRUE

3 Pretraining Tasks

3.1 Summary

Task	Train+	Train-	Hold+	Hold-	Total
poker_has_pair	20	20	20	20	80
poker_three_of_kind	20	20	20	20	80
poker_flush	20	20	20	20	80
poker_same_color	20	20	20	20	80
poker_two_suits	20	20	20	20	80
poker_straight	20	20	20	20	80
bj_sum_even	20	20	20	20	80
bj_sum_odd	20	20	20	20	80
rummy_run_3	20	20	20	20	80
rummy_set_3	20	20	20	20	80
rummy_all_different	20	20	20	20	80
rummy_three_ranks	20	20	20	20	80
sol_alternating	20	20	20	20	80
sol_v_shape	20	20	20	20	80
sol_ascending	20	20	20	20	80
sol_sorted	20	20	20	20	80
sol_same_suit_seq	20	20	20	20	80
simple_first_red	20	20	20	20	80
simple_last_black	20	20	20	20	80
simple_ends_suit	20	20	20	20	80
simple_ends_color	20	20	20	20	80
simple_ends_rank	20	20	20	20	80
simple_has_spade	20	20	20	20	80
simple_has_heart	20	20	20	20	80
simple_all_even	20	20	20	20	80
simple_all_odd	20	20	20	20	80
simple_diverse_ranks	20	20	20	20	80
simple_two_suits	20	20	20	20	80
sym_suits_palindrome	20	20	20	20	80
sym_periodic_colors	20	20	20	20	80
sym_ranks_palindrome	20	20	20	20	80
sym_halves_suits	20	20	20	20	80
count_two_red	20	20	20	20	80
count_more_red	20	20	20	20	80
count_three_suit	20	20	20	20	80
count_majority_color	20	20	20	20	80
comp_left_half_uniform_color	20	20	20	20	80
comp_right_half_has_pair	20	20	20	20	80
comp_skip1_same_color	20	20	20	20	80
comp_shift2_color	20	20	20	20	80
comp_first_two_same_suit	20	20	20	20	80
comp_last_two_same_color	20	20	20	20	80
comp_binary_suit_alt	20	20	20	20	80
comp_color_pairs	20	20	20	20	80

3.2 Detailed Task Examples

3.2.1 poker_has_pair

Family: poker, Difficulty: 1

Training Examples (first 6):

Hand	Label
NIN-C, NIN-H, FIV-C, FOU-D, FOU-S, TEN-S	TRUE
TEN-D, THR-S, TWO-C, EIG-S, THR-C, ACE-S	TRUE
ACE-S, SEV-H, NIN-S, JAC-H, FIV-C, QUE-H	FALSE
KIN-H, FOU-S, FIV-S, FIV-C, TWO-D, TWO-S	TRUE
TEN-H, KIN-C, QUE-D, FOU-C, JAC-D, ACE-D	FALSE
TWO-D, FIV-C, ACE-H, FOU-S, SEV-S, JAC-S	FALSE
Holdout Examples (first 4):	
Hand	Label
ACE-S, TEN-S, THR-S, JAC-H, KIN-D, EIG-D	FALSE
THR-C, ACE-D, KIN-D, FIV-S, SEV-C, EIG-H	FALSE
ACE-C, TWO-D, FIV-C, QUE-D, TEN-H, EIG-H	FALSE
THR-S, THR-H, JAC-D, KIN-D, THR-S, SIX-S	TRUE

3.2.2 poker_three_of_kind

Family: poker, Difficulty: 2

Training Examples (first 6):

Hand	Label
TEN-C, TEN-H, QUE-C, FIV-S, TWO-C, QUE-C	FALSE
KIN-D, SIX-H, TWO-S, TWO-D, TWO-C, NIN-H	TRUE
NIN-S, ACE-C, THR-S, ACE-S, ACE-C, FOU-S	TRUE
NIN-C, FIV-D, QUE-S, NIN-C, FIV-D, FOU-S	FALSE
NIN-C, NIN-D, FOU-D, TEN-D, TEN-D, FIV-C	FALSE
TWO-S, SEV-C, QUE-S, SEV-D, QUE-H, SEV-H	TRUE
Holdout Examples (first 4):	
Hand	Label
THR-D, NIN-H, FOU-C, NIN-D, EIG-D, NIN-S	TRUE
KIN-C, TEN-D, SIX-D, KIN-D, KIN-S, QUE-C	TRUE
FOU-C, ACE-D, EIG-C, SIX-C, EIG-H, KIN-H	FALSE
JAC-H, TEN-D, THR-C, THR-D, SIX-D, THR-S	TRUE

3.2.3 poker_flush

Family: poker, Difficulty: 1

Training Examples (first 6):

Hand	Label
NIN-C, JAC-C, NIN-C, FIV-C, EIG-C, SEV-C	TRUE
QUE-C, NIN-S, FIV-S, SEV-S, THR-S, EIG-S	FALSE
TEN-C, THR-C, QUE-S, SEV-C, FIV-C, NIN-C	FALSE
FOU-D, ACE-D, SIX-D, ACE-D, THR-D, NIN-D	TRUE
TWO-D, TEN-D, JAC-D, TWO-C, TWO-D, ACE-D	FALSE
EIG-S, ACE-S, FOU-C, JAC-S, THR-S, TWO-S	FALSE
Holdout Examples (first 4):	
Hand	Label
TWO-C, THR-C, TWO-C, SEV-C, EIG-C, THR-C	TRUE
THR-S, EIG-S, NIN-S, TEN-S, ACE-S, TWO-S	TRUE
JAC-H, TEN-D, THR-C, KIN-H, TWO-S, EIG-C	FALSE
SEV-H, EIG-C, QUE-S, JAC-S, SIX-S, TWO-C	FALSE

3.2.4 poker_same_color

Family: poker, Difficulty: 1

Training Examples (first 6):

Hand	Label
KIN-C, FOU-C, SIX-H, TWO-S, FOU-C, SEV-S	FALSE
THR-S, SEV-C, ACE-C, TEN-S, KIN-C, FIV-H	FALSE
JAC-C, SIX-S, TWO-C, ACE-S, EIG-S, NIN-C	TRUE
NIN-H, FOU-D, QUE-H, FOU-H, EIG-H, QUE-H	TRUE
KIN-H, THR-C, TEN-S, FIV-C, QUE-C, SEV-C	FALSE
EIG-H, THR-D, JAC-D, TWO-D, KIN-D, SEV-H	TRUE

Holdout Examples (first 4):

Hand	Label
EIG-C, SIX-H, FOU-C, KIN-C, FOU-S, SIX-H	FALSE
THR-S, JAC-C, KIN-S, TWO-C, FOU-D, KIN-H	FALSE
FIV-D, TEN-H, SEV-D, SIX-D, FIV-D, EIG-C	FALSE
JAC-S, QUE-D, SIX-D, JAC-H, SIX-D, SIX-C	FALSE

3.2.5 poker_two_suits

Family: poker, Difficulty: 1

Training Examples (first 6):

Hand	Label
ACE-C, TEN-S, NIN-H, QUE-S, KIN-S, THR-C	TRUE
SIX-C, TWO-C, EIG-C, FIV-C, FOU-C, TEN-C	FALSE
ACE-C, SEV-C, QUE-C, THR-C, QUE-C, KIN-C	FALSE
FOU-H, NIN-S, KIN-D, EIG-C, ACE-H, SIX-S	TRUE
JAC-C, JAC-C, EIG-C, JAC-C, FIV-C, TWO-C	FALSE
SIX-C, FOU-C, ACE-C, JAC-C, JAC-C, SIX-C	FALSE

Holdout Examples (first 4):

Hand	Label
QUE-S, SEV-S, SIX-S, NIN-S, TEN-S, THR-S	FALSE
ACE-C, QUE-C, TEN-H, EIG-C, FIV-C, TWO-D	TRUE
SIX-C, SIX-C, FIV-C, NIN-C, ACE-C, TEN-C	FALSE
QUE-S, TWO-S, EIG-H, TWO-H, QUE-H, SIX-S	TRUE

3.2.6 poker_straight

Family: poker, Difficulty: 2

Training Examples (first 6):

Hand	Label
QUE-S, ACE-H, KIN-C, QUE-C, SIX-C, TEN-S	FALSE
KIN-H, QUE-H, JAC-S, QUE-S, KIN-D, ACE-D	TRUE
SEV-D, QUE-D, TEN-C, JAC-D, QUE-S, QUE-D	FALSE
TEN-H, NIN-S, QUE-H, JAC-H, KIN-D, TEN-D	TRUE
SEV-H, FIV-S, EIG-D, FOU-D, SIX-D, SEV-D	TRUE
EIG-H, NIN-H, JAC-C, KIN-C, TEN-H, QUE-D	TRUE

Holdout Examples (first 4):

Hand	Label
TEN-H, JAC-D, TEN-S, QUE-S, NIN-C, JAC-S	TRUE
EIG-D, SEV-S, SIX-D, SEV-C, NIN-S, FIV-S	TRUE
NIN-H, EIG-S, QUE-H, QUE-C, ACE-D, TEN-D	FALSE
EIG-D, NIN-C, FIV-H, FIV-S, SEV-D, SIX-D	TRUE

3.2.7 bj_sum_even

Family: blackjack, Difficulty: 1

Training Examples (first 6):

Hand	Label
ACE-H, THR-H, JAC-H, KIN-C, EIG-C, EIG-S	FALSE
SIX-C, SEV-S, KIN-D, QUE-C, NIN-D, ACE-C	FALSE
JAC-S, SIX-D, THR-D, JAC-C, FOU-C, QUE-C	FALSE
QUE-D, THR-D, JAC-C, QUE-S, SEV-S, JAC-S	TRUE
KIN-H, JAC-H, TWO-H, THR-S, KIN-D, SIX-D	TRUE
FIV-D, SIX-D, KIN-C, THR-H, FIV-C, FOU-H	TRUE

Holdout Examples (first 4):

Hand	Label
THR-H, EIG-D, EIG-C, JAC-H, THR-H, FIV-D	TRUE
SIX-S, NIN-S, FOU-D, FOU-C, TEN-D, NIN-C	TRUE
THR-S, TWO-C, FIV-C, TEN-C, NIN-D, EIG-S	FALSE
NIN-D, KIN-C, EIG-C, TWO-C, FOU-H, THR-D	FALSE

3.2.8 bj_sum_odd

Family: blackjack, Difficulty: 1

Training Examples (first 6):

Hand	Label
ACE-C, QUE-H, KIN-C, TWO-D, EIG-H, FOU-S	TRUE
JAC-S, SIX-D, EIG-C, SEV-H, TEN-S, EIG-D	FALSE
SIX-H, EIG-D, JAC-C, FIV-S, NIN-S, SEV-D	FALSE
TEN-C, QUE-D, QUE-C, QUE-H, TEN-S, THR-D	TRUE
NIN-H, QUE-H, JAC-D, TEN-H, TWO-C, TWO-S	FALSE
TEN-C, SIX-S, TWO-S, FIV-S, KIN-C, NIN-S	TRUE

Holdout Examples (first 4):

Hand	Label
FIV-D, FIV-S, SEV-H, TEN-H, TWO-H, KIN-D	FALSE
SEV-C, TEN-S, THR-C, SIX-C, SIX-S, ACE-S	FALSE
TWO-C, TWO-D, EIG-D, THR-C, ACE-S, THR-H	FALSE
KIN-S, FOU-S, FOU-D, SEV-H, SEV-C, FOU-H	TRUE

3.2.9 rummy_run_3

Family: rummy, Difficulty: 2

Training Examples (first 6):

Hand	Label
FIV-C, NIN-S, SEV-H, SEV-S, THR-C, FIV-D	FALSE
THR-D, SEV-S, SEV-H, FOU-S, TEN-H, FIV-C	TRUE
JAC-D, SIX-D, ACE-C, QUE-C, TEN-H, FIV-H	TRUE
SIX-D, QUE-H, NIN-D, KIN-H, SIX-H, QUE-C	FALSE
FOU-S, ACE-C, JAC-H, EIG-C, THR-C, EIG-C	FALSE
FOU-S, FOU-D, TEN-S, KIN-C, TEN-S, NIN-S	FALSE

Holdout Examples (first 4):

Hand	Label
SEV-H, EIG-S, FOU-C, KIN-C, EIG-S, KIN-S	FALSE
JAC-C, TEN-C, ACE-S, QUE-S, ACE-H, JAC-C	TRUE
EIG-C, ACE-C, NIN-H, TWO-S, JAC-D, SEV-C	TRUE
EIG-H, FIV-D, KIN-S, ACE-D, SEV-H, NIN-S	TRUE

3.2.10 rummy_set_3**Family:** rummy, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
THR-H, TWO-H, FIV-C, FIV-H, TWO-H, EIG-H	FALSE
TWO-H, FIV-D, EIG-D, EIG-H, EIG-H, SEV-H	TRUE
KIN-H, TEN-C, FIV-D, FIV-H, FIV-H, SEV-D	TRUE
QUE-D, ACE-C, TWO-S, TEN-D, FOU-D, ACE-C	FALSE
TWO-C, JAC-S, TWO-H, FIV-C, SIX-D, TWO-H	TRUE
FIV-H, EIG-D, EIG-D, FIV-C, NIN-C, FIV-S	TRUE

Holdout Examples (first 4):

Hand	Label
JAC-S, FIV-C, TWO-H, TWO-C, EIG-H, EIG-C	FALSE
KIN-S, JAC-D, JAC-S, JAC-C, ACE-S, THR-S	TRUE
KIN-S, SIX-D, QUE-H, FOU-H, KIN-S, NIN-D	FALSE
FOU-D, KIN-D, SIX-S, KIN-H, FOU-H, KIN-C	TRUE

3.2.11 rummy_all_different**Family:** rummy, **Difficulty:** 1**Training Examples (first 6):**

Hand	Label
SEV-C, QUE-S, QUE-S, KIN-S, THR-D, NIN-S	FALSE
KIN-D, ACE-H, TEN-C, SIX-D, FOU-C, KIN-D	FALSE
KIN-S, THR-C, FOU-H, SIX-D, ACE-S, QUE-H	TRUE
EIG-S, JAC-D, TEN-D, TWO-S, ACE-C, KIN-H	TRUE
THR-C, TWO-S, TWO-C, FOU-C, SIX-S, KIN-H	FALSE
KIN-D, TEN-S, FOU-H, KIN-D, QUE-H, FIV-D	FALSE

Holdout Examples (first 4):

Hand	Label
FOU-D, FIV-S, SIX-C, KIN-S, EIG-H, SEV-D	TRUE
NIN-C, FOU-C, TWO-H, SIX-C, JAC-H, FIV-H	TRUE
TWO-D, TWO-S, SIX-C, NIN-C, JAC-H, THR-D	FALSE
FIV-S, KIN-C, NIN-C, NIN-D, NIN-D, TEN-C	FALSE

3.2.12 rummy_three_ranks**Family:** rummy, **Difficulty:** 1**Training Examples (first 6):**

Hand	Label
FOU-H, SEV-S, FOU-D, SEV-S, JAC-D, FOU-H	TRUE
TWO-D, TWO-H, TWO-D, JAC-S, ACE-S, JAC-C	TRUE
NIN-C, NIN-H, SEV-H, SEV-C, FIV-H, TWO-H	FALSE
KIN-D, KIN-D, NIN-D, FIV-H, FIV-S, FIV-S	TRUE
SIX-H, SEV-D, SEV-C, TEN-C, THR-D, THR-C	FALSE
SIX-S, NIN-S, QUE-H, TEN-D, NIN-D, SIX-H	FALSE

Holdout Examples (first 4):

Hand	Label
EIG-H, JAC-H, JAC-C, QUE-D, EIG-C, EIG-S	TRUE
SEV-D, KIN-H, KIN-S, KIN-H, FOU-D, FOU-H	TRUE
KIN-D, FOU-S, TWO-H, FOU-D, TWO-D, FOU-H	TRUE
SEV-H, QUE-D, QUE-H, FOU-D, KIN-C, EIG-H	FALSE

3.2.13 sol_alternating

Family: solitaire, Difficulty: 2

Training Examples (first 6):

Hand	Label
QUE-H, SEV-C, ACE-D, FIV-C, ACE-D, TWO-S	TRUE
QUE-S, SIX-D, FOU-C, TEN-S, ACE-C, EIG-H	FALSE
THR-C, TEN-S, SEV-C, THR-D, THR-S, NIN-D	FALSE
EIG-H, JAC-C, ACE-C, QUE-C, EIG-D, QUE-S	FALSE
JAC-C, TEN-D, ACE-H, QUE-D, FIV-S, SEV-H	FALSE
EIG-S, NIN-H, THR-S, FIV-D, FOU-S, QUE-D	TRUE

Holdout Examples (first 4):

Hand	Label
JAC-H, THR-S, QUE-H, QUE-C, FIV-H, FOU-D	FALSE
THR-S, SIX-D, THR-D, FIV-D, FIV-H, QUE-H	FALSE
FIV-C, ACE-C, TEN-S, JAC-S, NIN-C, KIN-H	FALSE
JAC-H, SEV-C, SIX-D, ACE-C, TEN-D, QUE-S	TRUE

3.2.14 sol_v_shape

Family: solitaire, Difficulty: 2

Training Examples (first 6):

Hand	Label
ACE-H, TEN-S, ACE-S, FOU-H, SIX-H, TEN-H	FALSE
ACE-C, ACE-H, ACE-H, FIV-D, SEV-C, KIN-C	TRUE
SEV-D, FIV-S, TWO-H, TWO-H, SIX-H, TWO-S	FALSE
KIN-D, QUE-C, EIG-D, TWO-C, SIX-C, NIN-H	TRUE
KIN-H, FOU-S, JAC-C, FOU-C, SIX-H, TEN-C	FALSE
SEV-C, THR-D, THR-C, TWO-H, FOU-S, TWO-C	FALSE

Holdout Examples (first 4):

Hand	Label
JAC-C, TEN-S, TWO-C, TWO-S, FOU-S, SEV-D	TRUE
JAC-D, FIV-D, SEV-D, KIN-C, JAC-C, SIX-S	FALSE
TEN-S, NIN-S, FIV-S, FOU-C, FIV-D, SIX-S	TRUE
ACE-C, TEN-C, NIN-S, THR-C, FIV-S, SEV-C	TRUE

3.2.15 sol_ascending

Family: solitaire, Difficulty: 2

Training Examples (first 6):

Hand	Label
TWO-D, FOU-C, SEV-S, NIN-C, JAC-H, QUE-D	TRUE
TWO-S, THR-H, FOU-D, SIX-C, JAC-S, KIN-S	TRUE
FOU-H, SEV-H, EIG-S, NIN-H, JAC-H, QUE-S	TRUE
THR-H, EIG-D, NIN-C, TEN-H, JAC-C, ACE-H	TRUE
THR-C, FOU-H, EIG-S, JAC-H, KIN-D, ACE-H	TRUE
TWO-S, FIV-H, SIX-S, TEN-D, QUE-S, KIN-H	TRUE

Holdout Examples (first 4):

Hand	Label
JAC-S, SIX-D, KIN-S, SIX-C, SEV-H, TWO-H	FALSE
FIV-H, SIX-S, SEV-S, NIN-C, QUE-D, ACE-H	TRUE
THR-H, SIX-C, QUE-H, SEV-H, ACE-D, KIN-H	FALSE
FOU-S, FIV-D, SEV-H, EIG-D, NIN-H, ACE-C	TRUE

3.2.16 sol_sorted**Family:** solitaire, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
FOU-S, FOU-C, NIN-S, TEN-H, QUE-S, ACE-C	TRUE
TWO-H, FOU-C, FOU-H, FOU-H, EIG-C, JAC-C	TRUE
THR-H, FIV-S, EIG-C, QUE-C, KIN-H, ACE-H	TRUE
QUE-S, SIX-D, SIX-S, SIX-S, EIG-H, NIN-H	FALSE
TWO-C, TWO-D, FIV-C, SIX-S, KIN-H, QUE-H	FALSE
TWO-C, FOU-D, SIX-D, SIX-C, FOU-D, ACE-S	FALSE

Holdout Examples (first 4):

Hand	Label
TEN-H, THR-S, TEN-H, JAC-C, FIV-C, ACE-D	FALSE
TEN-C, QUE-D, QUE-H, QUE-S, QUE-H, ACE-H	TRUE
SEV-S, NIN-S, NIN-D, NIN-C, NIN-D, ACE-C	TRUE
FOU-D, TEN-C, SEV-C, JAC-S, TEN-S, SEV-D	FALSE

3.2.17 sol_same_suit_seq**Family:** solitaire, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
FIV-S, TWO-S, SIX-S, SIX-D, THR-S, ACE-S	FALSE
THR-S, FOU-S, THR-S, TEN-S, SIX-S, NIN-S	TRUE
EIG-H, SIX-H, FIV-H, NIN-H, SIX-H, JAC-H	TRUE
SEV-H, TWO-H, SIX-H, TWO-H, THR-H, ACE-H	TRUE
THR-S, SEV-S, FOU-S, JAC-S, SIX-S, QUE-S	TRUE
SIX-C, ACE-C, QUE-S, EIG-C, ACE-C, SEV-C	FALSE

Holdout Examples (first 4):

Hand	Label
TEN-H, SIX-H, FIV-H, SIX-H, KIN-H, SIX-H	TRUE
ACE-C, FOU-C, FIV-C, KIN-C, THR-C, QUE-C	TRUE
THR-D, NIN-D, TEN-D, FOU-D, SEV-D, EIG-D	TRUE
FIV-H, NIN-H, EIG-H, TEN-H, FIV-H, EIG-H	TRUE

3.2.18 simple_first_red**Family:** simple, **Difficulty:** 1**Training Examples (first 6):**

Hand	Label
NIN-D, THR-S, EIG-C, SIX-H, TWO-H, KIN-H	TRUE
SEV-D, EIG-H, TWO-D, QUE-D, SEV-S, KIN-H	TRUE
THR-S, TEN-D, THR-D, FOU-S, SEV-H, SIX-H	FALSE
TWO-D, EIG-C, TWO-S, TEN-S, JAC-D, FOU-C	TRUE
ACE-C, SEV-D, TWO-D, FOU-D, FIV-D, THR-D	FALSE
KIN-D, TEN-D, EIG-D, NIN-C, NIN-S, NIN-H	TRUE

Holdout Examples (first 4):

Hand	Label
TEN-H, QUE-D, EIG-C, FOU-H, SEV-D, ACE-D	TRUE
SIX-C, TEN-H, FOU-C, FOU-H, TEN-H, NIN-D	FALSE
SEV-H, QUE-S, QUE-C, TWO-C, SEV-S, TEN-S	TRUE
SIX-D, EIG-S, SIX-H, SEV-C, EIG-H, KIN-D	TRUE

3.2.19 simple_last_black

Family: simple, Difficulty: 1

Training Examples (first 6):

Hand	Label
TWO-H, FOU-H, ACE-D, FOU-D, SEV-C, KIN-H	FALSE
EIG-S, EIG-S, TEN-D, QUE-C, THR-H, SEV-D	FALSE
THR-D, NIN-H, SIX-D, EIG-D, EIG-C, FIV-C	TRUE
SIX-D, JAC-C, FOU-D, TEN-H, FOU-H, TWO-H	FALSE
TWO-C, JAC-C, SIX-S, TEN-C, SEV-C, THR-H	FALSE
THR-S, TEN-H, QUE-H, FOU-H, ACE-H, JAC-H	FALSE

Holdout Examples (first 4):

Hand	Label
TEN-C, QUE-H, SIX-C, FOU-S, NIN-H, QUE-H	FALSE
TEN-C, QUE-S, KIN-C, FOU-H, ACE-C, FIV-H	FALSE
QUE-D, EIG-S, QUE-S, TEN-D, QUE-H, TWO-C	TRUE
SEV-H, FIV-S, JAC-H, TEN-C, SEV-S, NIN-S	TRUE

3.2.20 simple_ends_suit

Family: simple, Difficulty: 1

Training Examples (first 6):

Hand	Label
NIN-S, FIV-S, ACE-H, TEN-S, SEV-S, NIN-D	FALSE
FOU-H, QUE-D, THR-H, FIV-D, NIN-S, ACE-D	FALSE
FOU-H, ACE-D, NIN-S, FOU-H, KIN-H, FOU-C	FALSE
JAC-D, ACE-S, EIG-S, QUE-H, NIN-H, EIG-D	TRUE
QUE-C, TWO-S, SEV-H, TWO-S, NIN-H, SIX-C	TRUE
QUE-H, KIN-H, NIN-H, JAC-C, JAC-C, TWO-S	FALSE

Holdout Examples (first 4):

Hand	Label
EIG-C, TEN-H, JAC-D, SEV-C, EIG-S, FIV-C	TRUE
FIV-S, ACE-C, JAC-D, SEV-D, SEV-S, FOU-S	TRUE
QUE-D, FOU-S, SIX-H, TEN-S, TWO-D, QUE-D	TRUE
SEV-S, TWO-S, SIX-H, SIX-D, THR-H, EIG-S	TRUE

3.2.21 simple_ends_color

Family: simple, Difficulty: 1

Training Examples (first 6):

Hand	Label
FIV-C, SIX-C, QUE-S, SEV-C, SIX-H, KIN-H	FALSE
QUE-C, SEV-C, THR-H, TEN-D, SEV-C, TWO-C	TRUE
TWO-D, THR-D, SIX-C, THR-C, TWO-D, FOU-S	FALSE
FOU-C, SEV-C, TWO-S, SIX-C, TWO-S, THR-H	FALSE
KIN-D, THR-D, JAC-D, TWO-D, FOU-S, FOU-H	TRUE
TEN-S, JAC-D, TEN-C, SIX-S, FIV-S, KIN-S	TRUE

Holdout Examples (first 4):

Hand	Label
TWO-D, FOU-S, SEV-D, FOU-D, SEV-C, NIN-D	TRUE
THR-S, FOU-H, ACE-S, TEN-D, TEN-C, NIN-H	FALSE
FOU-C, FOU-C, TEN-D, NIN-S, THR-C, FIV-C	TRUE
KIN-H, FIV-D, KIN-C, THR-H, TEN-H, QUE-H	TRUE

3.2.22 simple_ends_rank

Family: simple, Difficulty: 1

Training Examples (first 6):

Hand	Label
QUE-H, SEV-D, EIG-D, NIN-S, KIN-S, NIN-D	FALSE
TWO-C, FOU-S, EIG-S, TEN-S, SIX-S, KIN-S	FALSE
NIN-D, ACE-D, THR-H, THR-S, TEN-H, NIN-C	TRUE
FIV-H, SEV-D, TEN-S, FIV-H, SEV-C, TWO-S	FALSE
QUE-S, FOU-H, FOU-H, FIV-D, NIN-H, FOU-D	FALSE
JAC-S, THR-C, EIG-H, KIN-H, TWO-S, TWO-H	FALSE

Holdout Examples (first 4):

Hand	Label
KIN-C, ACE-H, TWO-C, KIN-D, JAC-C, SEV-D	FALSE
TWO-D, NIN-D, THR-C, SEV-C, TWO-S, TWO-H	TRUE
THR-C, EIG-H, SIX-D, TEN-H, QUE-D, FIV-H	FALSE
JAC-C, ACE-C, JAC-D, SIX-C, FIV-H, SEV-S	FALSE

3.2.23 simple_has_spade

Family: simple, Difficulty: 1

Training Examples (first 6):

Hand	Label
THR-H, FIV-D, SEV-S, FIV-C, EIG-H, SIX-S	TRUE
JAC-D, FOU-D, THR-C, TEN-D, FOU-H, TEN-D	FALSE
THR-C, TEN-H, SIX-C, NIN-D, QUE-D, FOU-H	FALSE
TWO-D, SEV-D, NIN-C, NIN-H, ACE-D, SIX-C	FALSE
TEN-C, TWO-C, FOU-C, QUE-H, TWO-C, SEV-H	FALSE
ACE-H, ACE-C, SEV-D, FIV-D, EIG-C, KIN-D	FALSE

Holdout Examples (first 4):

Hand	Label
SEV-D, TEN-D, SIX-D, JAC-C, THR-H, THR-H	FALSE
TEN-C, FIV-H, ACE-C, FOU-H, TEN-S, TWO-D	TRUE
QUE-H, FIV-H, TEN-H, SIX-D, FIV-C, ACE-C	FALSE
FOU-C, THR-H, FIV-S, ACE-S, KIN-D, KIN-S	TRUE

3.2.24 simple_has_heart

Family: simple, Difficulty: 1

Training Examples (first 6):

Hand	Label
FIV-D, QUE-S, SIX-C, QUE-S, SIX-C, FIV-C	FALSE
ACE-D, SEV-C, TWO-C, NIN-S, KIN-D, QUE-D	FALSE
JAC-C, QUE-S, KIN-D, TEN-D, SEV-C, NIN-D	FALSE
JAC-H, NIN-D, TWO-S, JAC-H, SIX-C, TWO-C	TRUE
TEN-S, SIX-D, SEV-C, QUE-D, SEV-D, EIG-D	FALSE
FOU-C, EIG-C, FIV-H, FOU-H, THR-H, TEN-D	TRUE

Holdout Examples (first 4):

Hand	Label
NIN-D, FOU-H, SIX-D, FOU-C, SEV-D, TEN-S	TRUE
EIG-S, THR-H, TEN-S, TWO-D, EIG-D, NIN-H	TRUE
FOU-H, FOU-C, NIN-D, SIX-H, KIN-H, QUE-S	TRUE
KIN-D, TEN-H, FOU-D, SIX-H, FOU-S, TWO-D	TRUE

3.2.25 simple_all_even

Family: simple, Difficulty: 2

Training Examples (first 6):

Hand	Label
QUE-S, FOU-H, QUE-D, TEN-D, TWO-D, FOU-H	TRUE
FOU-S, TWO-D, SIX-H, FOU-S, TEN-H, FOU-D	TRUE
FOU-S, SIX-H, EIG-H, FOU-D, EIG-S, EIG-C	TRUE
FOU-S, EIG-S, NIN-H, EIG-D, TWO-S, TWO-H	FALSE
QUE-C, TWO-H, SIX-S, QUE-S, TWO-H, TEN-D	TRUE
ACE-C, TEN-C, JAC-H, TWO-C, ACE-H, TEN-C	FALSE

Holdout Examples (first 4):

Hand	Label
SEV-C, SIX-C, FIV-H, KIN-H, SEV-D, SIX-H	FALSE
TWO-D, SEV-D, JAC-C, FOU-S, EIG-H, THR-H	FALSE
QUE-H, TWO-H, KIN-H, THR-C, EIG-C, TWO-C	FALSE
TEN-H, EIG-S, TEN-S, EIG-D, SIX-H, TEN-S	TRUE

3.2.26 simple_all_odd

Family: simple, Difficulty: 2

Training Examples (first 6):

Hand	Label
KIN-S, SIX-S, KIN-C, SEV-C, FIV-H, SEV-H	FALSE
SEV-H, FIV-D, NIN-D, KIN-D, THR-H, SEV-H	TRUE
SEV-D, THR-D, THR-S, THR-D, SEV-H, QUE-S	FALSE
THR-C, JAC-C, SEV-C, SEV-H, KIN-D, JAC-D	TRUE
KIN-H, KIN-S, SEV-H, THR-D, SEV-H, NIN-D	TRUE
SEV-S, NIN-D, FIV-S, FIV-C, KIN-H, NIN-H	TRUE

Holdout Examples (first 4):

Hand	Label
JAC-C, ACE-H, TWO-D, QUE-C, NIN-H, FOU-S	FALSE
TEN-C, FOU-C, FOU-S, ACE-H, NIN-D, EIG-C	FALSE
THR-S, NIN-D, JAC-H, ACE-D, FIV-S, KIN-H	FALSE
SEV-H, NIN-C, KIN-H, KIN-S, FIV-H, NIN-H	TRUE

3.2.27 simple_diverse_ranks

Family: simple, Difficulty: 1

Training Examples (first 6):

Hand	Label
SEV-H, FOU-C, THR-H, KIN-H, QUE-S, EIG-H	TRUE
SEV-H, QUE-S, THR-C, ACE-C, TWO-C, TWO-S	TRUE
FOU-S, FOU-D, TEN-H, TEN-D, QUE-C, QUE-H	FALSE
FOU-S, THR-C, FOU-S, THR-H, TWO-D, TWO-H	FALSE
FOU-C, SEV-D, FOU-D, FOU-H, FOU-C, KIN-H	FALSE
SIX-C, EIG-H, FOU-H, JAC-S, NIN-H, THR-D	TRUE

Holdout Examples (first 4):

Hand	Label
ACE-H, JAC-D, SIX-S, NIN-C, NIN-S, SEV-H	TRUE
THR-S, TWO-D, SEV-H, THR-S, ACE-H, EIG-D	TRUE
KIN-S, KIN-C, NIN-S, QUE-S, NIN-D, NIN-S	FALSE
EIG-D, EIG-C, FIV-D, JAC-H, JAC-S, EIG-S	FALSE

3.2.28 simple_two_suits

Family: simple, Difficulty: 1

Training Examples (first 6):

Hand	Label
NIN-C, EIG-H, TWO-H, TEN-C, JAC-C, THR-S	FALSE
NIN-H, SEV-S, FIV-H, NIN-H, NIN-H, FIV-H	TRUE
TWO-C, SIX-D, THR-C, QUE-D, TWO-S, FIV-D	FALSE
QUE-H, SEV-D, SIX-D, NIN-D, KIN-D, FOU-H	TRUE
SEV-S, EIG-C, SIX-D, EIG-D, TWO-C, KIN-C	FALSE
JAC-C, FOU-C, FIV-D, KIN-C, TWO-D, THR-D	TRUE

Holdout Examples (first 4):

Hand	Label
KIN-C, QUE-C, ACE-C, NIN-D, QUE-C, THR-H	FALSE
TWO-D, SIX-D, FOU-S, TWO-S, NIN-S, SIX-H	FALSE
QUE-D, EIG-H, SEV-H, JAC-S, ACE-C, JAC-C	FALSE
TEN-C, SEV-D, TWO-C, FIV-D, NIN-D, FOU-D	TRUE

3.2.29 sym_suits_palindrome

Family: symmetry, Difficulty: 2

Training Examples (first 6):

Hand	Label
ACE-H, ACE-D, SEV-C, ACE-C, TEN-S, QUE-H	FALSE
QUE-S, JAC-S, ACE-C, TWO-C, TWO-S, QUE-S	TRUE
FOU-S, SIX-C, FOU-H, QUE-H, FIV-S, TWO-S	FALSE
QUE-C, EIG-C, TEN-D, NIN-H, EIG-C, SEV-C	FALSE
ACE-H, ACE-S, QUE-C, SEV-C, JAC-S, JAC-H	TRUE
NIN-D, JAC-D, SEV-H, TEN-H, EIG-D, EIG-D	TRUE

Holdout Examples (first 4):

Hand	Label
JAC-D, QUE-H, JAC-S, QUE-S, JAC-H, ACE-D	TRUE
SIX-C, SEV-S, ACE-H, KIN-H, SEV-S, QUE-C	TRUE
FOU-D, KIN-H, TEN-D, THR-D, SIX-H, EIG-D	TRUE
ACE-C, SIX-D, TWO-S, TWO-H, ACE-D, SIX-H	FALSE

3.2.30 sym_periodic_colors

Family: symmetry, Difficulty: 2

Training Examples (first 6):

Hand	Label
ACE-D, FIV-C, EIG-D, SEV-H, JAC-C, KIN-S	FALSE
FIV-S, TEN-H, SEV-S, TEN-D, KIN-C, SIX-H	TRUE
FIV-H, TEN-C, THR-S, FIV-D, TEN-C, KIN-S	TRUE
SIX-D, ACE-C, THR-C, JAC-D, FOU-C, FIV-C	TRUE
JAC-C, QUE-D, FOU-C, SEV-S, SIX-D, FIV-D	FALSE
QUE-C, FOU-S, SEV-C, TWO-C, SEV-H, KIN-S	FALSE

Holdout Examples (first 4):

Hand	Label
TEN-H, SEV-D, SEV-D, EIG-S, EIG-C, TWO-S	FALSE
THR-D, JAC-D, FIV-C, NIN-C, FIV-S, ACE-D	FALSE
JAC-H, NIN-C, KIN-S, SIX-D, SEV-C, TWO-S	TRUE
SEV-S, EIG-C, FIV-C, FIV-C, JAC-C, EIG-C	TRUE

3.2.31 sym_ranks_palindrome**Family:** symmetry, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
SEV-C, EIG-H, JAC-C, JAC-H, EIG-C, SEV-C	TRUE
THR-D, TEN-S, SEV-S, SEV-C, FIV-C, THR-S	FALSE
JAC-C, TWO-D, THR-C, THR-H, NIN-D, JAC-S	FALSE
QUE-D, ACE-D, TEN-C, TEN-H, ACE-D, SEV-C	FALSE
TEN-S, FIV-D, EIG-D, EIG-S, FIV-D, TEN-C	TRUE
THR-C, SIX-D, SEV-S, SEV-H, SIX-H, THR-D	TRUE

Holdout Examples (first 4):

Hand	Label
TEN-D, NIN-S, TWO-H, TWO-C, NIN-S, TEN-S	TRUE
FIV-H, ACE-H, JAC-S, TEN-C, TEN-H, KIN-H	FALSE
TWO-S, THR-D, EIG-C, EIG-H, THR-C, TWO-C	TRUE
ACE-C, FIV-C, QUE-D, QUE-D, FIV-C, ACE-H	TRUE

3.2.32 sym_halves_suits**Family:** symmetry, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
KIN-C, EIG-D, NIN-S, EIG-H, FOU-D, NIN-S	FALSE
SEV-C, SEV-D, SIX-H, SEV-S, SEV-D, NIN-H	FALSE
JAC-S, SEV-C, KIN-C, ACE-S, SEV-C, THR-C	TRUE
JAC-H, FOU-S, SEV-C, TWO-H, THR-S, SEV-C	TRUE
TEN-H, QUE-H, SIX-D, SEV-C, EIG-H, TWO-D	FALSE
QUE-D, TWO-C, KIN-D, TEN-D, FIV-D, TWO-D	FALSE

Holdout Examples (first 4):

Hand	Label
QUE-C, SIX-D, TWO-H, EIG-C, SEV-D, SEV-H	TRUE
QUE-S, NIN-D, SIX-H, NIN-D, JAC-H, THR-H	FALSE
JAC-H, TEN-S, TEN-C, QUE-H, FIV-S, QUE-C	TRUE
TWO-C, FIV-H, TWO-C, FOU-C, QUE-H, NIN-C	TRUE

3.2.33 count_two_red**Family:** counting, **Difficulty:** 1**Training Examples (first 6):**

Hand	Label
FOU-C, QUE-S, SIX-S, FOU-D, FOU-D, NIN-C	TRUE
FIV-S, ACE-H, SIX-C, FIV-D, FIV-D, ACE-S	FALSE
ACE-H, JAC-S, JAC-S, KIN-S, FIV-S, SEV-S	FALSE
TEN-C, KIN-H, JAC-D, EIG-C, TWO-S, SEV-S	TRUE
SEV-C, JAC-D, SEV-C, SIX-S, JAC-H, FOU-D	FALSE
NIN-C, SEV-C, KIN-H, NIN-S, SEV-C, FIV-S	FALSE

Holdout Examples (first 4):

Hand	Label
SIX-C, NIN-D, EIG-C, SEV-C, SIX-D, EIG-C	TRUE
THR-S, JAC-H, KIN-H, SEV-H, JAC-D, EIG-H	FALSE
QUE-C, SIX-H, TEN-D, SEV-H, FIV-S, SIX-D	FALSE
SEV-D, SIX-S, KIN-S, QUE-C, JAC-D, FIV-H	FALSE

3.2.34 count_more_red**Family:** counting, **Difficulty:** 1**Training Examples (first 6):**

Hand	Label
NIN-S, KIN-C, JAC-H, ACE-H, SEV-D, ACE-S	FALSE
EIG-S, THR-D, SEV-C, TWO-D, TEN-D, THR-H	TRUE
ACE-H, EIG-H, NIN-D, KIN-D, QUE-H, FOU-S	TRUE
THR-H, FIV-H, SEV-D, NIN-S, SIX-H, TEN-D	TRUE
NIN-H, TEN-H, JAC-H, KIN-D, ACE-C, NIN-S	TRUE
QUE-D, TWO-C, ACE-S, EIG-C, KIN-H, FIV-D	FALSE

Holdout Examples (first 4):

Hand	Label
SIX-D, ACE-C, FIV-H, SEV-D, FOU-H, KIN-S	TRUE
NIN-H, QUE-C, QUE-H, EIG-S, NIN-D, FIV-S	FALSE
THR-S, ACE-D, KIN-C, QUE-D, QUE-H, NIN-S	FALSE
JAC-S, KIN-S, FOU-S, THR-D, KIN-D, SEV-D	FALSE

3.2.35 count_three_suit**Family:** counting, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
JAC-H, TEN-S, TEN-C, QUE-H, EIG-H, ACE-D	TRUE
THR-D, TWO-S, ACE-C, TWO-H, THR-D, FIV-S	FALSE
JAC-C, JAC-S, FOU-D, TEN-D, SIX-C, THR-D	TRUE
ACE-S, JAC-H, QUE-C, ACE-H, SIX-C, NIN-H	TRUE
QUE-C, ACE-D, SIX-H, ACE-D, TEN-S, SEV-C	FALSE
NIN-S, EIG-H, ACE-C, QUE-H, SEV-C, JAC-C	TRUE

Holdout Examples (first 4):

Hand	Label
KIN-C, THR-S, FOU-D, FIV-D, ACE-H, TEN-H	FALSE
THR-H, NIN-C, JAC-H, FIV-D, SEV-H, SIX-D	TRUE
TEN-S, SIX-D, NIN-D, TWO-H, SEV-H, FOU-C	FALSE
THR-D, SEV-S, SIX-H, QUE-S, SEV-C, TWO-D	FALSE

3.2.36 count_majority_color**Family:** counting, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
FOU-D, EIG-S, SIX-H, TWO-C, SIX-H, TEN-C	FALSE
THR-S, SIX-H, EIG-S, NIN-S, EIG-H, JAC-D	FALSE
SIX-S, FOU-C, TEN-D, JAC-D, FOU-D, QUE-D	TRUE
QUE-H, SIX-C, SEV-C, TEN-H, FIV-S, SIX-D	FALSE
SIX-H, NIN-C, FOU-S, THR-C, FIV-H, FOU-H	FALSE
FOU-D, JAC-H, JAC-H, NIN-S, NIN-C, NIN-H	TRUE

Holdout Examples (first 4):

Hand	Label
FOU-S, SEV-C, TEN-S, QUE-S, JAC-H, SEV-C	TRUE
THR-D, SIX-S, SIX-D, SIX-D, SIX-H, QUE-H	TRUE
THR-H, TWO-S, QUE-C, TWO-D, TWO-S, SEV-C	TRUE
KIN-C, SEV-C, SIX-S, EIG-D, ACE-S, NIN-C	TRUE

3.2.37 comp_left_half_uniform_color

Family: compositional, Difficulty: 2

Training Examples (first 6):

Hand	Label
NIN-C, THR-H, SIX-S, JAC-C, ACE-H, ACE-S	FALSE
JAC-S, THR-S, TEN-S, JAC-H, FIV-H, TEN-D	TRUE
QUE-C, FOU-S, JAC-S, TWO-D, THR-D, SEV-H	TRUE
ACE-D, NIN-H, SIX-H, THR-S, FOU-D, EIG-C	TRUE
SIX-H, TEN-D, KIN-D, JAC-S, JAC-D, SEV-S	TRUE
KIN-C, SEV-C, NIN-C, KIN-H, FOU-H, TWO-S	TRUE

Holdout Examples (first 4):

Hand	Label
THR-D, FIV-H, ACE-H, ACE-C, FIV-C, THR-H	TRUE
KIN-D, FOU-H, JAC-C, JAC-S, TWO-D, FIV-H	FALSE
FIV-C, SIX-C, THR-S, FOU-C, KIN-C, SEV-D	TRUE
SEV-D, KIN-S, QUE-D, JAC-H, EIG-D, TEN-D	FALSE

3.2.38 comp_right_half_has_pair

Family: compositional, Difficulty: 2

Training Examples (first 6):

Hand	Label
THR-S, KIN-D, SIX-C, QUE-S, KIN-S, TEN-H	FALSE
QUE-D, NIN-H, SEV-H, QUE-C, FIV-S, FIV-S	TRUE
TWO-C, NIN-S, SIX-S, FOU-H, FOU-H, SEV-D	TRUE
QUE-C, TWO-C, EIG-S, NIN-D, TEN-C, EIG-D	FALSE
NIN-D, TEN-D, JAC-C, KIN-C, TEN-S, TWO-C	FALSE
JAC-S, ACE-D, QUE-H, SEV-C, JAC-H, THR-C	FALSE

Holdout Examples (first 4):

Hand	Label
FIV-H, ACE-C, TWO-H, JAC-C, SIX-C, SIX-H	TRUE
TWO-D, TEN-D, TEN-C, QUE-C, JAC-C, SIX-H	FALSE
SIX-H, JAC-C, SIX-D, TWO-C, SEV-D, SEV-H	TRUE
JAC-H, KIN-H, SEV-D, EIG-S, EIG-S, FOU-S	TRUE

3.2.39 comp_skip1_same_color

Family: compositional, Difficulty: 2

Training Examples (first 6):

Hand	Label
FIV-C, THR-S, FIV-S, EIG-C, EIG-C, KIN-S	TRUE
NIN-C, FOU-D, SIX-C, FIV-H, EIG-H, JAC-H	FALSE
JAC-C, SEV-S, NIN-S, KIN-H, TWO-C, SIX-H	FALSE
FOU-C, TWO-H, FIV-C, FOU-D, NIN-S, JAC-H	TRUE
JAC-H, EIG-D, NIN-D, FOU-D, NIN-D, SEV-D	TRUE
FOU-H, KIN-D, EIG-D, SIX-H, JAC-D, TWO-S	FALSE

Holdout Examples (first 4):

Hand	Label
ACE-S, JAC-D, SIX-C, FIV-H, SEV-S, NIN-H	TRUE
SIX-C, FOU-H, NIN-S, KIN-H, SIX-S, TWO-D	TRUE
JAC-H, SEV-H, TEN-D, KIN-D, TWO-H, NIN-H	TRUE
FIV-C, ACE-C, SEV-C, THR-C, TWO-C, FIV-S	TRUE

3.2.40 comp_shift2_color**Family:** compositional, **Difficulty:** 2**Training Examples (first 6):**

Hand	Label
ACE-D, FOU-C, TWO-C, NIN-C, FOU-D, NIN-C	FALSE
THR-H, NIN-D, ACE-H, TWO-S, FIV-D, NIN-D	FALSE
NIN-D, SIX-D, NIN-H, FIV-D, FOU-D, FIV-S	FALSE
TEN-H, FOU-H, QUE-H, NIN-H, ACE-D, EIG-S	FALSE
SIX-S, TEN-S, NIN-D, QUE-S, EIG-S, SIX-C	FALSE
SIX-H, KIN-H, EIG-H, KIN-D, JAC-D, SEV-D	TRUE

Holdout Examples (first 4):

Hand	Label
FIV-S, SEV-H, NIN-D, SIX-S, TEN-C, EIG-H	FALSE
EIG-S, EIG-S, SIX-S, SEV-S, EIG-S, THR-S	TRUE
EIG-H, NIN-H, JAC-C, THR-C, QUE-D, SIX-C	FALSE
QUE-H, ACE-H, ACE-C, TEN-C, JAC-D, SEV-C	FALSE

3.2.41 comp_first_two_same_suit**Family:** compositional, **Difficulty:** 1**Training Examples (first 6):**

Hand	Label
QUE-C, TEN-H, THR-S, THR-C, JAC-C, TWO-S	FALSE
FOU-C, EIG-C, JAC-H, SIX-H, ACE-H, ACE-H	TRUE
JAC-S, TWO-S, KIN-D, SIX-H, KIN-H, FIV-S	TRUE
FIV-S, EIG-S, SEV-H, KIN-C, EIG-D, TEN-H	TRUE
SIX-H, QUE-C, SEV-C, THR-S, KIN-D, SIX-H	FALSE
TWO-H, KIN-H, THR-C, ACE-D, FOU-S, NIN-H	TRUE

Holdout Examples (first 4):

Hand	Label
FOU-H, EIG-H, EIG-D, FIV-C, SEV-H, FOU-C	TRUE
EIG-D, ACE-D, SEV-D, TEN-D, FIV-C, FOU-D	TRUE
SEV-C, THR-C, THR-D, EIG-S, THR-S, THR-S	TRUE
FOU-S, QUE-S, EIG-S, TWO-D, ACE-H, EIG-H	TRUE

3.2.42 comp_last_two_same_color**Family:** compositional, **Difficulty:** 1**Training Examples (first 6):**

Hand	Label
QUE-H, EIG-S, KIN-C, JAC-H, QUE-C, TWO-D	FALSE
NIN-C, FOU-H, JAC-D, ACE-C, JAC-S, NIN-C	TRUE
FOU-S, QUE-C, TWO-H, ACE-C, THR-D, FIV-C	FALSE
FIV-S, EIG-H, FOU-H, SEV-D, JAC-D, SIX-H	TRUE
EIG-C, FIV-D, QUE-S, JAC-C, FOU-H, QUE-H	TRUE
FOU-D, FOU-S, TEN-D, QUE-S, TEN-S, TWO-H	FALSE

Holdout Examples (first 4):

Hand	Label
QUE-S, JAC-D, EIG-C, SEV-D, SEV-H, QUE-D	TRUE
SEV-H, THR-D, FOU-D, KIN-D, ACE-C, NIN-H	FALSE
QUE-D, NIN-S, KIN-H, ACE-H, THR-S, FOU-H	FALSE
KIN-H, TEN-S, TEN-C, FOU-H, QUE-D, SIX-C	FALSE

3.2.43 comp_binary_suit_alt

Family: compositional, **Difficulty:** 2

Training Examples (first 6):

Hand	Label
EIG-S, TWO-D, KIN-S, ACE-D, NIN-S, ACE-D	TRUE
FOU-D, SIX-C, KIN-S, TWO-C, SIX-D, FIV-C	FALSE
EIG-H, THR-S, FOU-H, ACE-S, TWO-H, SIX-S	TRUE
SEV-C, SIX-D, JAC-H, SEV-D, ACE-H, TEN-D	FALSE
QUE-C, TWO-D, KIN-C, NIN-D, TEN-C, SEV-D	TRUE
TWO-C, KIN-S, QUE-C, JAC-S, THR-C, SEV-S	TRUE

Holdout Examples (first 4):

Hand	Label
SEV-C, FOU-S, QUE-C, FIV-S, THR-S, FIV-D	FALSE
TWO-C, JAC-C, THR-S, TEN-H, FOU-D, JAC-H	FALSE
FIV-D, SEV-D, EIG-C, KIN-S, SEV-C, QUE-S	FALSE
QUE-C, NIN-C, JAC-D, TWO-D, SIX-S, TWO-D	FALSE

3.2.44 comp_color_pairs

Family: compositional, **Difficulty:** 2

Training Examples (first 6):

Hand	Label
QUE-C, SEV-S, SEV-H, TEN-H, SIX-D, ACE-H	TRUE
JAC-C, THR-S, TEN-S, FIV-S, FIV-C, ACE-S	TRUE
KIN-S, ACE-C, THR-C, QUE-S, THR-S, ACE-C	TRUE
EIG-C, TWO-S, JAC-D, EIG-C, QUE-C, NIN-C	FALSE
NIN-S, THR-S, SEV-S, EIG-C, EIG-D, TWO-H	TRUE
JAC-D, TWO-D, JAC-S, KIN-H, KIN-H, ACE-H	FALSE

Holdout Examples (first 4):

Hand	Label
ACE-S, ACE-S, EIG-H, FIV-H, TEN-H, FIV-H	TRUE
KIN-D, EIG-H, QUE-S, FIV-C, FOU-C, FOU-C	TRUE
THR-S, KIN-H, ACE-C, KIN-S, SIX-D, NIN-C	FALSE
EIG-H, EIG-D, TWO-D, TWO-H, KIN-H, JAC-C	FALSE

4 Generation Metadata

4.1 Catalogue

- Generated: 2026-01-01T18:30:36.904477
- Seed: 42
- Source: rules/catalogue.py - ALL RULES
- Tasks attempted: 45
- Tasks generated: 44
- Failures: 1

Failed rules:

- Shift_half_plus_two: Could not find enough positive examples for Shift_half_plus_two. Found 45, neede...

Configuration:

- Training positives: 20
- Training negatives: 20
- Seed positives (for near-miss): 20
- Holdout positives: 20
- Holdout negatives: 20
- Near-miss enabled: True
- Max sampling attempts: 200,000

4.2 Pretraining

- Generated: 2026-01-01T18:30:45.604238
- Seed: 1000
- Source: rules/pretraining rules.py - PRETRAINING RULES
- Tasks attempted: 44
- Tasks generated: 44
- Failures: 0

Configuration:

- Training positives: 20
- Training negatives: 20
- Seed positives (for near-miss): 20
- Holdout positives: 20
- Holdout negatives: 20
- Near-miss enabled: True
- Max sampling attempts: 200,000